

Cuaderno de Prácticas

Álgebra

Grado en Ingeniería Informática

CURSO 2024/25

Datos personales

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Grupo de teoría : A

Grupo de prácticas :

Grafos

Práctica 4

Código

```
In[1]:= (*Matriz de adyacencia*)
MATRIZADYACENCIA[W_,F_] := Module[{k,i,j,CONTADORi,nuevoF},
nuevoF = {};
Do[AppendTo[nuevoF,Position[W,F[[CONTADORi]][[1]][[1]] -> Position[W,F[[CONTADORi]][[2]][[1]][[1]]],
,{CONTADORi,Length[F]}];
matrizadyacencia = Table[0,{i,Length[W]},{j,Length[W]}];
Do[matrizadyacencia[[nuevoF[[k,1]],nuevoF[[k,2]]] = 1;
matrizadyacencia[[nuevoF[[k,2]],nuevoF[[k,1]]] = 1;
,{k,Length[F]}];
matrizadyacencia];

(*Matriz de incidencia*)
MATRIZINCIDENCIA[W_,F_] := Module[{i,j,nuevoF,CONTADORi},
nuevoF = {};
Do[AppendTo[nuevoF,Position[W,F[[CONTADORi]][[1]][[1]] -> Position[W,F[[CONTADORi]][[2]][[1]][[1]]],
,{CONTADORi,Length[F]}];
matrizincidencia = Table[0,{i,Length[W]},{j,Length[F]}];
Do[Do[
```

```

matrizincidencia[[nuevoF[[j]][i],j]]=1;
,{i,2}]],{j,Length[F]}}];
matrizincidencia
];

(*Grado*)
GRADO[n_,F_]:=Module[{i},
grado=0;
Do[If[Intersection[{F[[i]][1],F[[i]][2]},{n]}=={n},grado++],{i,Length[F]}}];
grado
]
(*Regular*)
REGULAR[matrizadyacencia_]:=Module[{i,j},
grado=Sum[matrizadyacencia[[1,j]],{j,Dimensions[matrizadyacencia][[1]]}];
regular=True;
Do[
If[grado!=Sum[matrizadyacencia[[i,j]],{j,1,Dimensions[matrizadyacencia][[1]]}],regular=False;
Break[[]];
,{i,2,Dimensions[matrizadyacencia][[1]]}];
regular
];
REGULAR2[W_,F_]:=Module[{i,j},
grado=GRADO[W[[1]],F];
regular=True;
Do[
If[grado!=GRADO[W[[i]],F],regular=False];
,{i,2,Length[W]}}];
If[regular,Print["El grafo es ",grado,"-regular"],Print["El grafo no es regular"]];
];
(*Completo*)
COMPLETO[matrizadyacencia_]:=
(Sum[MatrixPower[matrizadyacencia,2][[i,i],
{i,Dimensions[matrizadyacencia][[1]]}]==
Dimensions[matrizadyacencia][[1]]*(Dimensions[matrizadyacencia][[1]]-1))

```

Out[2]=

```

In[8]:= (*Isomorfos*)
ISOMORFOS[A1_,A2_] := Module[{matricesP,i,j},
  Isomorfos=False;
  If[Dimensions[A1]==Dimensions[A2],
    matricesP=Permutations[IdentityMatrix[Dimensions[A1][[1]]]];
    Do[
      If[matricesP[[i]].A1.Transpose[matricesP[[i]]]==A2,
        Isomorfos=True;matrizP=matricesP[[i]];permutacion=i;Break[;;];
        ,{i,Length[matricesP]}];
    ];

    If[Isomorfos,
      Print["Son isomorfos, una matriz P es: ",MatrixForm[matrizP],
        ", una permutación es: ",
        MatrixForm[{Table[j,{j,Dimensions[A1][[1]]}],Permutations[Table[j,{j,Dimensions[A1][[1]]}]]
        [[permutacion]]}]]
      ,Print["No son isomorfos"]];
    ];

```

Out[8]=

```

In[10]:= (*Bipartito*)
IND[W1_,F_] := Module[{i},
  F1={};
  Do[If[Sort[Intersection[{F[[i][[1]],F[[i][[2]]}],W1]]==Sort[{F[[i][[1]],F[[i][[2]]}],AppendTo[F1,
  F[[i]]],{i,Length[F]}];
  F1
  ]
  BIPARTITO[W_,F_] := Module[{i,subconjuntos,j,fin},
    bipartito=False;
    i=1;
    fin=Quotient[Length[W],2];
    While[!bipartito && i<=fin,
      subconjuntos=KSubsets[W,i];
      j=1;
      While[!bipartito && j<=Length[subconjuntos],
        If[IND[subconjuntos[[j]],F]=={ },
          If[IND[Complement[W,subconjuntos[[j]]],F]=={ },
            bipartito=True;
            W1=subconjuntos[[j]];
            W2=Complement[W,subconjuntos[[j]]];
          ];
        ];
        j++;];
        i++;];
        If[bipartito,Print["Es bipartito: W1=",W1," y W2=",W2];
        If[Length[F]==Length[W1]*Length[W2],Print["Es bipartito completo: ", "K" Length[W1], Length[W2]
        ], Print["No es bipartito completo"]];
        ,Print["No es bipartito"]];
        ];

```

Ejercicio 5.29 Grafos I

- a) Sea W el conjunto formado por todas las cifras distintas que componen su DNI junto con todas las letras distintas de su primer apellido.
- b) Consideramos el grafo no orientado G_1 cuyo conjunto de vértices es W y donde dos vértices cualesquiera serán adyacentes si y sólo si son:
- I. Un número par y una vocal
 - II. Un número impar y una consonante
 - III. Dos números pares distintos
- c) Consideramos el grafo dirigido G_2 cuyo conjunto de vértices es W y donde el conjunto de flechas está formado por todas aquellas flechas que podemos considerar de la forma:
- I. Origen en un número impar y fin de una vocal
 - II. Origen en una vocal y fin en una constante
 - III. Origen en un número par y fin en un número impar

Se pide:

- i. Definir G_1 y G_2 directamente por sus conjuntos de vértices y lados o flechas

```
In[12]:= W={2,6,8,0,, "m", "a", "r", "t", "i", "n", "l", "u", "s"}; (*Comparten vértices ambos grafos*)
(*ARISTAS PRIMER GRAFO: par a vocal y pares distintos*)
F1={
  {0, "a"}, {0, "i"}, {0, "u"},
  {2, "a"}, {2, "i"}, {2, "u"},
  {6, "a"}, {6, "i"}, {6, "u"},
  {8, "a"}, {8, "i"}, {8, "u"},
  {2, 6}, {2, 8}, {2, 6}
};
(*ARISTAS SEGUNDO GRAFO: impar a vocal , vocal a consonante y par a impar *)
F2={
  {"a", "m"}, {"a", "r"}, {"a", "t"}, {"a", "n"}, {"a", "l"}, {"a", "s"},
  {"i", "m"}, {"i", "r"}, {"i", "t"}, {"i", "n"}, {"i", "l"}, {"i", "s"},
  {"u", "m"}, {"u", "r"}, {"u", "t"}, {"u", "n"}, {"u", "l"}, {"u", "s"}
};
```

- ii. Calcular las matrices de adyacencia de G_1 y G_2

```
In[15]:= "G1"
A1=MATRIZADYACENCIA[W,F1];
A1//MatrixForm
"G2"
A2=MATRIZADYACENCIA[W,F2];
A2//MatrixForm
```

Out[15]=

G1

Out[17]//MatrixForm=

[illegible]

Out[18]=

G2

Out[20]//MatrixForm=

[illegible]

iii. Calcular la matriz de incidencia de G_1

```
In[21]:= "G1"
I1=MATRIZINCIDENCIA[W,F1];
I1//MatrixForm
```

Out[21]=

G1

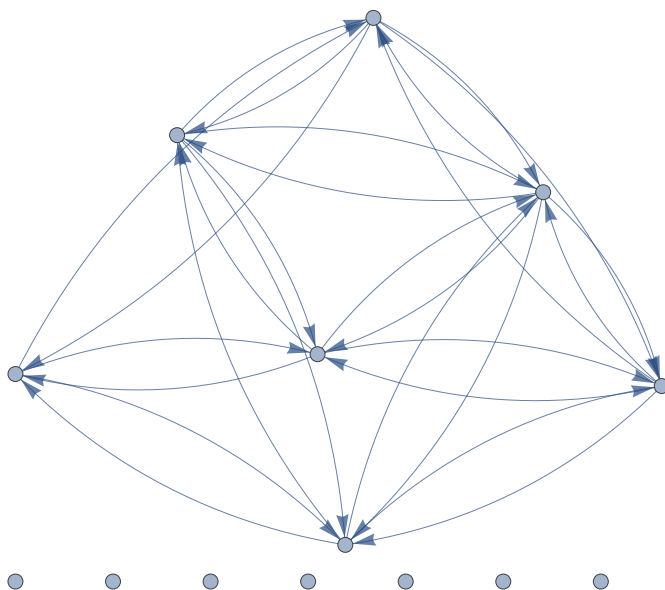
Out[23]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

iv. Representar gráficamente G_1 y G_2

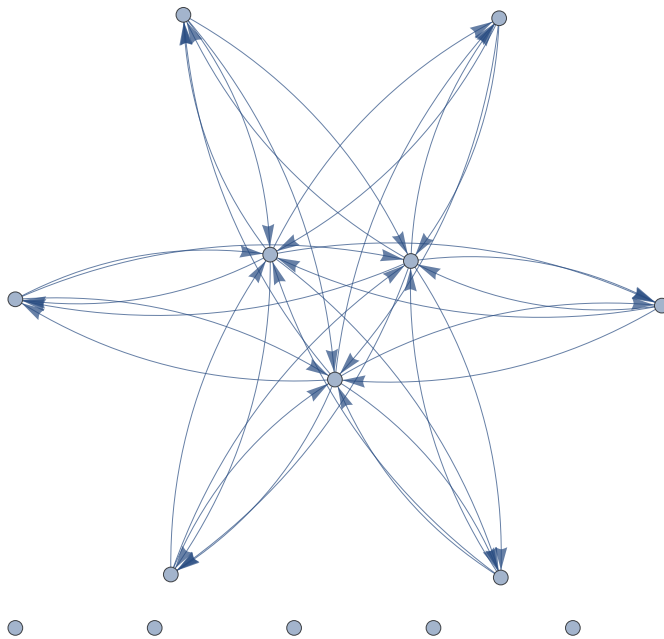
```
In[24]:= GraphPlot[A1,DirectedEdges→True]
```

Out[24]=



```
In[25]:= GraphPlot[A2, DirectedEdges→True]
```

```
Out[25]=
```



Ejercicio 2 Grafos I

Calcular conjunto de vértices y lados, matriz de adyacencia, matriz de incidencia y representación gráfica de los grafos del ejercicio 6.8 del manual de prácticas

- a) $G_1 = (W_1, F_1)$ con $W_1 = \{1, 2, 3, 4, 5, 6\}$ y $F_1 = \{1 \rightarrow 2, 3 \rightarrow 1, 4 \rightarrow 5, 6 \rightarrow 1\}$.
 b) G_2 con matriz de adyacencia:

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

- c) G_3 con matriz de incidencia:

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

a)

```

In[26]:= WEjer2GIa={1,2,3,4,5,6};
FEjer2GIa={1→2,3→1,4→5,6→1}

"MATRIZADYACENCIA"
AEjer2GIa=MATRIZADYACENCIA[WEjer2GIa,FEjer2GIa];
AEjer2GIa//MatrixForm
"MATRIZINCIDENCIA"
IEjer2GIa=MATRIZINCIDENCIA[WEjer2GIa,FEjer2GIa];
IEjer2GIa//MatrixForm
"Vertices y lados"
matrizadyacencia=AEjer2GIa;
W=Table[i,{i,Dimensions[matrizadyacencia][[1]]}]
F={};
Do[Do[If[matrizadyacencia[[i,j]]==1,AppendTo[F,i→j]]];,{i,Dimensions[matrizadyacencia][[1]]}];,
{ j,Dimensions[matrizadyacencia][[1]]}];
F
"Representacion gráfica"
GraphPlot[AEjer2GIa,DirectedEdges→True]

```

```

Out[27]=
{1 → 2, 3 → 1, 4 → 5, 6 → 1}

```

```

Out[28]=
MATRIZADYACENCIA

```

```

Out[30]//MatrixForm=

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$


```

```

Out[31]=
MATRIZINCIDENCIA

```

```

Out[33]//MatrixForm=

$$\begin{pmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$


```

```

Out[34]=
Vertices y lados

```

```

Out[36]=
{1, 2, 3, 4, 5, 6}

```

```

Out[39]=
{2 → 1, 3 → 1, 6 → 1, 1 → 2, 1 → 3, 5 → 4, 4 → 5, 1 → 6}

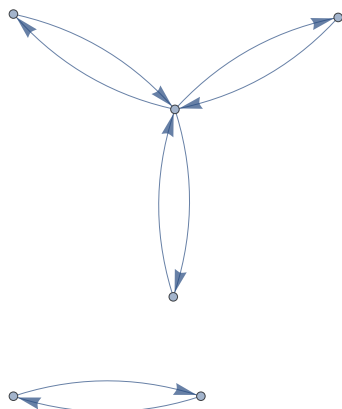
```

```

Out[40]=
Representacion gráfica

```


Out[41]=



b)

```

In[42]:= matrizadyacenciaEjer2GIb =  $\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$ ;

W=Table[i,{i,Dimensions[matrizadyacenciaEjer2GIb][[1]]}]
F={};
Do[Do[If[matrizadyacenciaEjer2GIb[[i,j]]==1,AppendTo[F,i->j]]],{i,Dimensions
[matrizadyacenciaEjer2GIb][[1]]}],{j,Dimensions[matrizadyacenciaEjer2GIb][[1]]}];
F

```

Out[43]=

{1, 2, 3, 4}

Out[46]=

{2 → 1, 4 → 1, 1 → 2, 3 → 2, 2 → 3, 4 → 3, 1 → 4, 3 → 4}

```

In[47]:= WEjer2GIb={1,2,3,4};
FEjer2GIb={2→1,4→1,1→2,3→2,2→3,4→3,1→4,3→4};
"MATRIZINCIDENCIA"
IEjer2GIb=MATRIZINCIDENCIA[WEjer2GIb,FEjer2GIb];
IEjer2GIb//MatrixForm
"Vertices y lados"
W=Table[i,{i,Dimensions[matrizadyacenciaEjer2GIb][[1]]}]
F={};
Do[Do[If[matrizadyacenciaEjer2GIb[[i,j]]==1,AppendTo[F,i→j]]],{i,Dimensions
[matrizadyacenciaEjer2GIb][[1]]}],{j,Dimensions[matrizadyacenciaEjer2GIb][[1]]}];
F
"Representacion gráfica"
GraphPlot[matrizadyacenciaEjer2GIb,DirectedEdges→True]

```

Out[49]=

MATRIZINCIDENCIA

Out[51]//MatrixForm=

$$\begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix}$$

Out[52]=

Vertices y lados

Out[53]=

{1, 2, 3, 4}

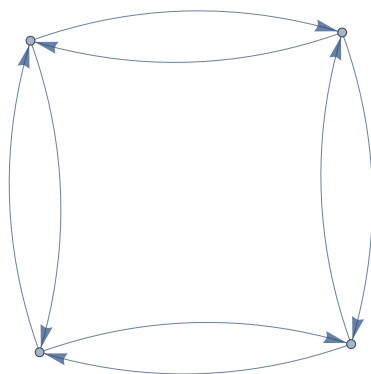
Out[56]=

{2 → 1, 4 → 1, 1 → 2, 3 → 2, 2 → 3, 4 → 3, 1 → 4, 3 → 4}

Out[57]=

Representacion gráfica

Out[58]=



c)

In[59]:=

$$\text{matrizincidenciaEjer2GIc} = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix};$$

```
W=Table[i,{i,Dimensions[matrizincidenciaEjer2GIc][[1]]}]
F={};
Do[
AppendTo[F,Position[Transpose[matrizincidenciaEjer2GIc][[i]],1][[1]][1]→Position[Transpose
[matrizincidenciaEjer2GIc][[i]],1][[2]][1]]
,{i,Dimensions[matrizincidenciaEjer2GIc][[2]]}];
F
```

Out[60]=

{1, 2, 3, 4, 5}

Out[63]=

{1 → 4, 1 → 5, 2 → 4, 2 → 5, 3 → 4, 3 → 5}

```

In[64]:= WEjer2GIC={1,2,3,4,5};
FEjer2GIC={1→4,1→5,2→4,2→5,3→4,3→5};
"MATRIZADYACENCIA"
AEjer2GIC=MATRIZADYACENCIA[WEjer2GIC,FEjer2GIC];
AEjer2GIC//MatrixForm
"Vertices y lados"
W=Table[i,{i,Dimensions[AEjer2GIC][[1]]}]
F={};
Do[Do[If[AEjer2GIC[[i,j]]==1,AppendTo[F,i→j]]],{i,Dimensions[AEjer2GIC][[1]]}],{j,Dimensions[AEjer2GIC][[1]]}];
F
"Representacion gráfica"
GraphPlot[AEjer2GIC,DirectedEdges→True]

```

Out[66]=

MATRIZADYACENCIA

Out[68]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \end{pmatrix}$$

Out[69]=

Vertices y lados

Out[70]=

{1, 2, 3, 4, 5}

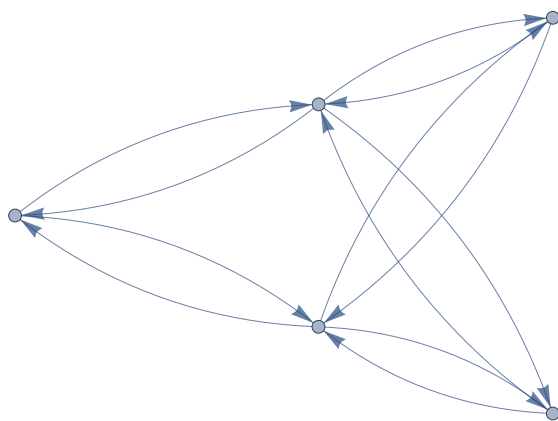
Out[73]=

{4 → 1, 5 → 1, 4 → 2, 5 → 2, 4 → 3, 5 → 3, 1 → 4, 2 → 4, 3 → 4, 1 → 5, 2 → 5, 3 → 5}

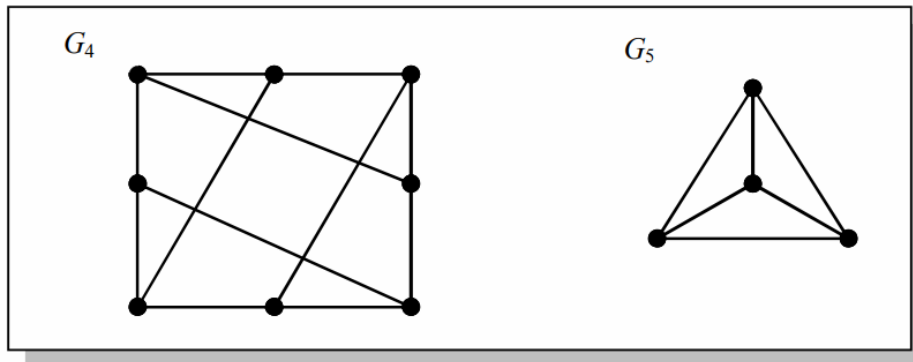
Out[74]=

Representacion gráfica

Out[75]=



d) Los grafos de la siguiente ilustración:



d)

```
In[76]:= "d"
WEjer2GIdg4={1,2,3,4,5,6,7,8};
FEjer2GIdg4={1→2,2→3,3→4,4→5,5→6,6→7,7→8,8→1,1→4,2→7,3→6,5→8}
"MATRIZADYACENCIA"
AEjer2GIdg4=MATRIZADYACENCIA[WEjer2GIdg4,FEjer2GIdg4];
AEjer2GIdg4//MatrixForm
"MATRIZINCIDENCIA"
IEjer2GIdg4=MATRIZINCIDENCIA[WEjer2GIdg4,FEjer2GIdg4];
IEjer2GIdg4//MatrixForm
"Vertices y lados"
matrizadyacencia=AEjer2GIdg4;
W=Table[i,{i,Dimensions[matrizadyacencia][[1]]}]
F={};
Do[Do[If[matrizadyacencia[[i,j]]==1,AppendTo[F,i→j]]],{i,Dimensions[matrizadyacencia][[1]]}],{j,Dimensions[matrizadyacencia][[1]]}];
F
"Representacion gráfica"
GraphPlot[AEjer2GIdg4,DirectedEdges→True]
```

Out[76]=

d

Out[78]=

{1 → 2, 2 → 3, 3 → 4, 4 → 5, 5 → 6, 6 → 7, 7 → 8, 8 → 1, 1 → 4, 2 → 7, 3 → 6, 5 → 8}

Out[79]=

MATRIZADYACENCIA

Out[81]//MatrixForm=

$$\begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Out[82]=

MATRIZINCIDENCIA

Out[84]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{pmatrix}$$

Out[85]=

Vertices y lados

Out[87]=

{1, 2, 3, 4, 5, 6, 7, 8}

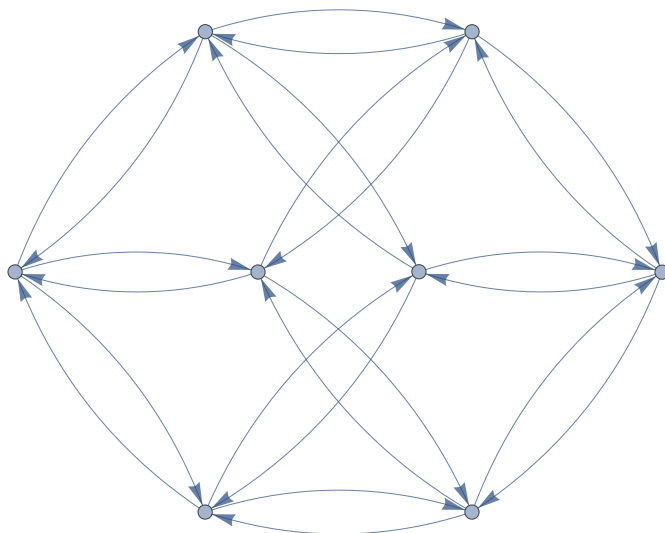
Out[90]=

{2 → 1, 4 → 1, 8 → 1, 1 → 2, 3 → 2, 7 → 2, 2 → 3, 4 → 3, 6 → 3, 1 → 4, 3 → 4, 5 → 4,
4 → 5, 6 → 5, 8 → 5, 3 → 6, 5 → 6, 7 → 6, 2 → 7, 6 → 7, 8 → 7, 1 → 8, 5 → 8, 7 → 8}

Out[91]=

Representacion gráfica

Out[92]=

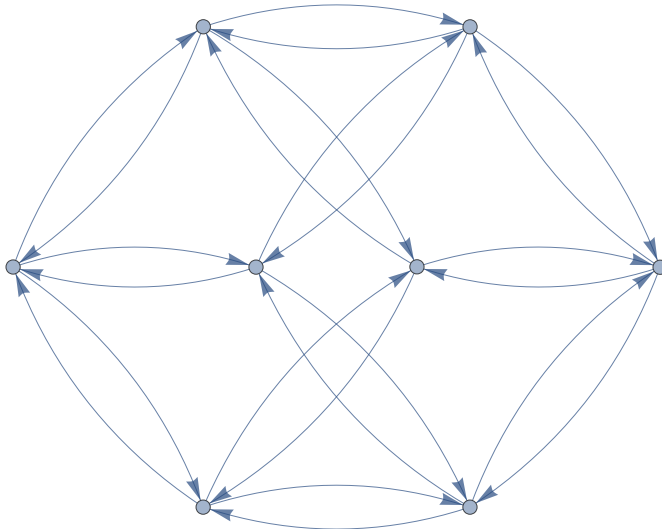


```
{2 → 1, 4 → 1, 8 → 1, 1 → 2, 3 → 2, 7 → 2, 2 → 3,
 4 → 3, 6 → 3, 1 → 4, 3 → 4, 5 → 4, 4 → 5, 6 → 5, 8 → 5,
 3 → 6, 5 → 6, 7 → 6, 2 → 7, 6 → 7, 8 → 7, 1 → 8, 5 → 8, 7 → 8}
```

Out[*]=

Representacion gráfica

Out[*]=



In[109]:=

```
WEjer2GIdg5={1,2,3,4};
FEjer2GIdg5={1→2,2→3,3→1,1→4,2→4,3→4}
"MATRIZADYACENCIA"
AEjer2GIdg5=MATRIZADYACENCIA[WEjer2GIdg5,FEjer2GIdg5];
AEjer2GIdg5//MatrixForm
"MATRIZINCIDENCIA"
IEjer2GIdg5=MATRIZINCIDENCIA[WEjer2GIdg5,FEjer2GIdg5];
IEjer2GIdg5//MatrixForm
"Vertices y lados"
matrizadyacencia=AEjer2GIdg5;
W=Table[i,{i,Dimensions[matrizadyacencia][[1]]}]
F={};
Do[Do[If[matrizadyacencia[[i,j]]==1,AppendTo[F,i→j]]],{i,Dimensions[matrizadyacencia][[1]]}];
,{j,Dimensions[matrizadyacencia][[1]]}];
F
"Representacion gráfica"
GraphPlot[AEjer2GIdg5,DirectedEdges→True]
```

Out[110]=

```
{1 → 2, 2 → 3, 3 → 1, 1 → 4, 2 → 4, 3 → 4}
```

Out[111]=

MATRIZADYACENCIA

Out[113]//MatrixForm=

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

Out[114]=

MATRIZINCIDENCIA

Out[116]//MatrixForm=

$$\begin{pmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{pmatrix}$$

Out[117]=

Vertices y lados

Out[119]=

{1, 2, 3, 4}

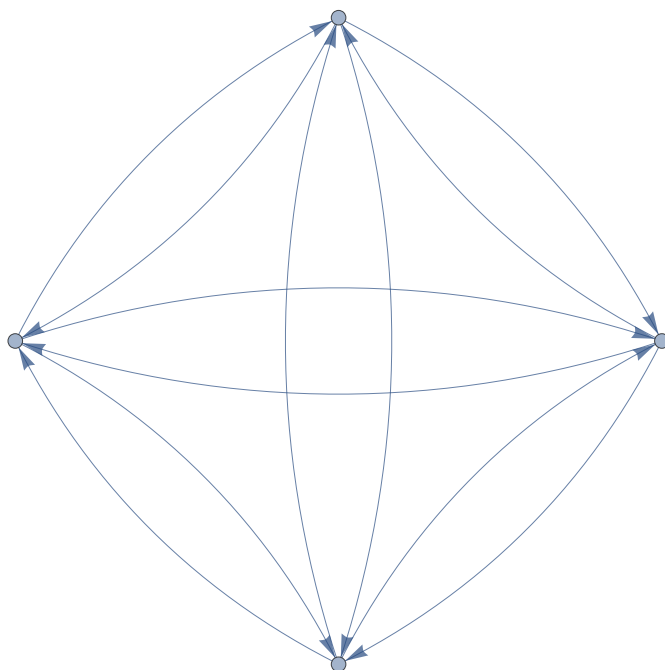
Out[122]=

{2 → 1, 3 → 1, 4 → 1, 1 → 2, 3 → 2, 4 → 2, 1 → 3, 2 → 3, 4 → 3, 1 → 4, 2 → 4, 3 → 4}

Out[123]=

Representacion gráfica

Out[124]=



Ejercicio 3 Grafos I

Ejercicio 5.4 del manual de prácticas

Sacar

a)

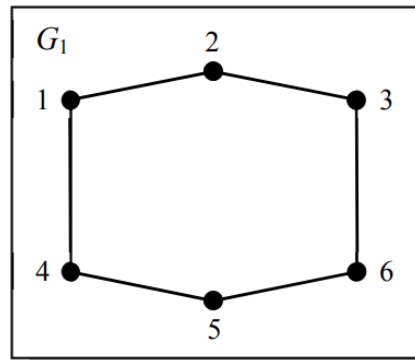


Ilustración 5.20.

b) G_2 , el grafo con matriz de adyacencia:

$$A_2 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

c)

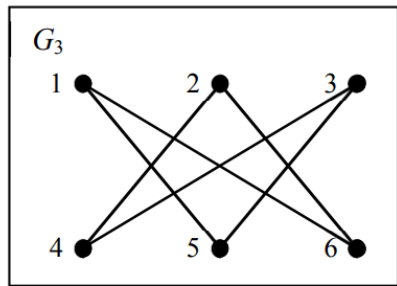


Ilustración 5.21.

d) G_4 , el grafo con matriz de adyacencia:

$$A_4 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & 0 \end{pmatrix}$$

e) G_5 , el grafo con matriz de incidencia:

$$B = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

f) $G_6 = (W, F)$ con $W = \{1, 2, 3, 4, 5, 6\}$ y $F = \{1 \rightarrow 2, 1 \rightarrow 3, 2 \rightarrow 3, 3 \rightarrow 5, 4 \rightarrow 5, 4 \rightarrow 6, 5 \rightarrow 6\}$

□

In[125]:=

```

matrizincidenciaEjer3GIc=

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \end{pmatrix};$$


W=Table[i,{i,Dimensions[matrizincidenciaEjer3GIc][[1]]}]
F={};
Do[
AppendTo[F,Position[Transpose[matrizincidenciaEjer3GIc][[i]],1][[1]][1]→Position[Transpose[matrizincidenciaEjer3GIc][[i]],1][[2]][1]]
,{i,Dimensions[matrizincidenciaEjer3GIc][[2]]}];
F

```

Out[126]=

```
{1, 2, 3, 4, 5, 6}
```

Out[129]=

```
{1 → 6, 2 → 5, 3 → 4, 1 → 2, 3 → 5, 4 → 6}
```

In[130]:=

```

"G1";
WEjer3GIIdg1={1,2,3,4,5,6};
FEjer3GIIdg1={{1,2},{2,3},{3,6},{6,5},{5,4},{4,2}};
"G3";
WEjer3GIIdg3={1,2,3,4,5,6};
FEjer3GIIdg3={{1,6},{1,5},{2,4},{2,6},{3,4},{3,5}};
"G5";
WEjer3GIIdg5={1,2,3,4,5,6};
FEjer3GIIdg5={1→6,2→5,3→4,1→2,3→5,4→6};
"G6";
WEjer3GIIdg6={1,2,3,4,5,6};
FEjer3GIIdg6={1→2, 1→3, 2→3, 3→5, 4→5,4→6, 5→6};

```

In[142]:=

```

"G1"
AG1=MATRIZADYACENCIA[WEjer3GIdg1,FEjer3GIdg1];
MatrixForm[AG1]
"G2"
AG2=
$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}$$

"G3"
AG3=MATRIZADYACENCIA[WEjer3GIdg3,FEjer3GIdg3];
MatrixForm[AG3]
"G4"
AG4=
$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & 0 \end{pmatrix}$$

"G5"
AG5=MATRIZADYACENCIA[WEjer3GIdg5,FEjer3GIdg5];
MatrixForm[AG5]
"G6"
AG6=MATRIZADYACENCIA[WEjer3GIdg6,FEjer3GIdg6];
MatrixForm[AG6]

```

Out[142]=

G1

Out[144]//MatrixForm=

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Out[145]=

G2

Out[146]=

```

{{0, 1, 0, 0, 0, 1}, {1, 0, 1, 0, 0, 0}, {0, 1, 0, 1, 0, 0},
{0, 0, 1, 0, 1, 0}, {0, 0, 0, 1, 0, 1}, {1, 0, 0, 0, 1, 0}}

```

Out[147]=

G3

Out[149]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Out[150]=

G4

Out[151]=

$\{\{0, 1, 0, 0, 0, 1\}, \{1, 0, 0, 0, 0, 1\}, \{0, 0, 0, 1, 0, 0\},$
 $\{0, 0, 1, 0, 1, 0\}, \{0, 0, 0, 1, 0, 1\}, \{1, 1, 0, 0, 1, 0\}\}$

Out[152]=

G5

Out[154]//MatrixForm=

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

Out[155]=

G6

Out[157]//MatrixForm=

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}$$

$\{\{0, 1, 0, 0, 0, 1\}, \{1, 0, 1, 0, 0, 0\}, \{0, 1, 0, 1, 0, 0\}, \{0, 0, 1, 0, 1, 0\},$
 $\{0, 0, 0, 1, 0, 1\}, \{1, 0, 0, 0, 1, 0\}\}$

Out[*]=

G3

Out[*]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Out[*]=

G4

```
{ {0, 1, 0, 0, 0, 1}, {1, 0, 0, 0, 0, 1}, {0, 0, 0, 1, 0, 0}, {0, 0, 1, 0, 1, 0},
  {0, 0, 0, 1, 0, 1}, {1, 1, 0, 0, 1, 0} }
```

```
Out[ ]=
```

G5

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

```
Out[ ]=
```

G6

```
Out[ ]//MatrixForm=
```

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}$$

```
In[158]:=
```

```
"G1 y G2"
ISOMORFOS [AG1,AG2]
"G1 y G3"
ISOMORFOS [AG1,AG3]
"G1 y G4"
ISOMORFOS [AG1,AG4]
"G1 y G5"
ISOMORFOS [AG1,AG5]
"G1 y G6"
ISOMORFOS [AG1,AG6]
"G2 y G3"
ISOMORFOS [AG2,AG3]
"G2 y G4"
ISOMORFOS [AG2,AG4]
"G2 y G5"
ISOMORFOS [AG2,AG5]
"G2 y G6"
ISOMORFOS [AG2,AG6]
"G3 y G4"
ISOMORFOS [AG3,AG4]
"G3 y G5"
ISOMORFOS [AG3,AG5]
"G3 y G6"
ISOMORFOS [AG3,AG6]
"G4 y G5"
ISOMORFOS [AG4,AG5]
"G4 y G6"
ISOMORFOS [AG4,AG6]
"G5 y G6"
ISOMORFOS [AG5,AG6]
```

Out[158]=

G1 y G2

No son isomorfos

Out[160]=

G1 y G3

No son isomorfos

Out[162]=

G1 y G4

No son isomorfos

Out[164]=

G1 y G5

No son isomorfos

Out[166]=

G1 y G6

No son isomorfos

Out[168]=

G2 y G3

Son isomorfos, una matriz P es:
$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 \end{pmatrix}, \text{ una permutaci3n es: } \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 3 & 5 & 4 & 6 & 2 \end{pmatrix}$$

Out[170]=

G2 y G4

No son isomorfos

Out[172]=

G2 y G5

Son isomorfos, una matriz P es:
$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \text{ una permutaci3n es: } \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 4 & 5 & 3 & 6 \end{pmatrix}$$

Out[174]=

G2 y G6

No son isomorfos

Out[176]=

G3 y G4

No son isomorfos

Out[178]=

G3 y G5

Son isomorfos, una matriz P es:
$$\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \text{ una permutaci3n es: } \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 5 & 4 & 2 & 3 & 6 \end{pmatrix}$$

```

Out[180]=
  G3 y G6

  No son isomorfos

Out[182]=
  G4 y G5

  No son isomorfos

Out[184]=
  G4 y G6

  No son isomorfos

Out[186]=
  G5 y G6

  No son isomorfos

```

Ejercicio 1: Grafos II

Determinar cuáles de los siguientes grafos son regulares, completos, bipartitos o bipartitos completos.

- a) $G_1 = (W_1, F_1)$ con $W_1 = \{1, 2, 3, 4, 5, 6\}$ y $F_1 = \{1 \rightarrow 2, 3 \rightarrow 1, 4 \rightarrow 5, 6 \rightarrow 1\}$.
 b) G_2 con matriz de adyacencia:

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

- c) G_3 con matriz de incidencia:

$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

a)

In[188]:=

```
WEjer2GIIa={1,2,3,4,5,6};
FEjer2GIIa={{1,2},{3,1},{4,5},{6,1}};
AEjer2GIIa=MATRIZADYACENCIA[WEjer2GIIa,FEjer2GIIa];
MatrixForm[AEjer2GIIa]
REGULAR2[WEjer2GIIa,FEjer2GIIa]
COMPLETO[AEjer2GIIa]
BIPARTITO[WEjer2GIIa,FEjer2GIIa]
```

Out[191]//MatrixForm=

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

El grafo no es regular

Out[193]=

False

Intersection: Nonatomic expression expected at position 2 in $\{1, 2\} \cap 1$. [i](#)

Intersection: Nonatomic expression expected at position 1 in $1 \cap \{1, 2\}$. [i](#)

Intersection: Nonatomic expression expected at position 2 in $\{3, 1\} \cap 1$. [i](#)

General: Further output of Intersection::normal will be suppressed during this calculation. [i](#)

Complement: Nonatomic expression expected at position 2 in Complement[{1, 2, 3, 4, 5, 6}, 1]. [i](#)

Intersection: Heads Complement and List at positions 2 and 1 are expected to be the same. [i](#)

Intersection: Heads List and Complement at positions 2 and 1 are expected to be the same. [i](#)

Intersection: Heads Complement and List at positions 2 and 1 are expected to be the same. [i](#)

General: Further output of Intersection::heads will be suppressed during this calculation. [i](#)

Complement: Nonatomic expression expected at position 2 in Complement[{1, 2, 3, 4, 5, 6}, 1]. [i](#)

Es bipartito: $W1=1$ y $W2=\text{Complement}[\{1, 2, 3, 4, 5, 6\}, 1]$

No es bipartito completo

In[195]:=

```
WEjer2GIIG2={1,2,3,4};
FEjer2GIIG2={1→2,2→3,1→4,3→4}
AEjer2GII2a=MATRIZADYACENCIA[WEjer2GIIG2,FEjer2GIIG2];
MatrixForm[AEjer2GII2a]
REGULAR2[WEjer2GIIG2,FEjer2GIIG2]
COMPLETO[AEjer2GII2a]
BIPARTITO[WEjer2GIIG2,FEjer2GIIG2]
```

Out[196]=

 $\{1 \rightarrow 2, 2 \rightarrow 3, 1 \rightarrow 4, 3 \rightarrow 4\}$

Out[198]//MatrixForm=

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

El grafo es 2-regular

Out[200]=

False

- ... **Intersection:** Nonatomic expression expected at position 2 in $\{1, 2\} \cap 1$. [i](#)
- ... **Intersection:** Nonatomic expression expected at position 1 in $1 \cap \{1, 2\}$. [i](#)
- ... **Intersection:** Nonatomic expression expected at position 2 in $\{2, 3\} \cap 1$. [i](#)
- ... **General:** Further output of Intersection::normal will be suppressed during this calculation. [i](#)
- ... **Complement:** Nonatomic expression expected at position 2 in Complement[{1, 2, 3, 4}, 1]. [i](#)
- ... **Intersection:** Heads Complement and List at positions 2 and 1 are expected to be the same. [i](#)
- ... **Intersection:** Heads List and Complement at positions 2 and 1 are expected to be the same. [i](#)
- ... **Intersection:** Heads Complement and List at positions 2 and 1 are expected to be the same. [i](#)
- ... **General:** Further output of Intersection::heads will be suppressed during this calculation. [i](#)
- ... **Complement:** Nonatomic expression expected at position 2 in Complement[{1, 2, 3, 4}, 1]. [i](#)

Es bipartito: W1=1 y W2=Complement[{1, 2, 3, 4}, 1]

No es bipartito completo

In[202]:=

```
WEjer2GIIG3={1,2,3,4,5};
FEjer2GIIG3={1→4,1→5,2→4,2→5,3→4,2→5};
AEjer2GII3a=MATRIZADYACENCIA[WEjer2GIIG3,FEjer2GIIG3];
MatrixForm[AEjer2GII3a]
REGULAR2[WEjer2GIIG3,FEjer2GIIG3]
COMPLETO[AEjer2GII3a]
BIPARTITO[WEjer2GIIG3,FEjer2GIIG3]
```

Out[205]//MatrixForm=

$$\begin{pmatrix} 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \end{pmatrix}$$

El grafo no es regular

Out[207]=

False

Intersection: Nonatomic expression expected at position 2 in $\{1, 4\} \cap 1$. [i](#)

Intersection: Nonatomic expression expected at position 1 in $1 \cap \{1, 4\}$. [i](#)

Intersection: Nonatomic expression expected at position 2 in $\{1, 5\} \cap 1$. [i](#)

General: Further output of Intersection::normal will be suppressed during this calculation. [i](#)

Complement: Nonatomic expression expected at position 2 in Complement[{1, 2, 3, 4, 5}, 1]. [i](#)

Intersection: Heads Complement and List at positions 2 and 1 are expected to be the same. [i](#)

Intersection: Heads List and Complement at positions 2 and 1 are expected to be the same. [i](#)

Intersection: Heads Complement and List at positions 2 and 1 are expected to be the same. [i](#)

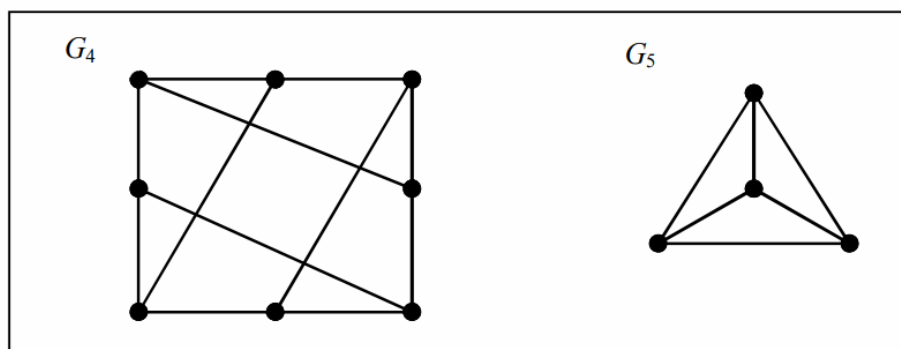
General: Further output of Intersection::heads will be suppressed during this calculation. [i](#)

Complement: Nonatomic expression expected at position 2 in Complement[{1, 2, 3, 4, 5}, 1]. [i](#)

Es bipartito: $W1=1$ y $W2=\text{Complement}[\{1, 2, 3, 4, 5\}, 1]$

No es bipartito completo

d) Los grafos de la siguiente ilustración:



In[209]:=

```
WEjer2GIIG4={1,2,3,4,5,6,7,8};
FEjer2GIIG4={{1,2},{2,3},{3,4},{4,5},{5,6},{6,7},{7,8},{1,4},{2,7},{3,6},{5,8}};
AEjer2GII4a=MATRIZADYACENCIA[WEjer2GIIG4,FEjer2GIIG4];
MatrixForm[AEjer2GII4a]
REGULAR2[WEjer2GIIG4,FEjer2GIIG4]
COMPLETO[AEjer2GII4a]
BIPARTITO[WEjer2GIIG4,FEjer2GIIG4]
```

Out[212]//MatrixForm=

$$\begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

El grafo no es regular

Out[214]=

False

- Intersection: Nonatomic expression expected at position 2 in {1, 2} ∩ 1. [i](#)
- Intersection: Nonatomic expression expected at position 1 in 1 ∩ {1, 2}. [i](#)
- Intersection: Nonatomic expression expected at position 2 in {2, 3} ∩ 1. [i](#)
- General: Further output of Intersection::normal will be suppressed during this calculation. [i](#)
- Complement: Nonatomic expression expected at position 2 in Complement[{1, 2, 3, 4, 5, 6, 7, 8}, 1]. [i](#)
- Intersection: Heads Complement and List at positions 2 and 1 are expected to be the same. [i](#)
- Intersection: Heads List and Complement at positions 2 and 1 are expected to be the same. [i](#)
- Intersection: Heads Complement and List at positions 2 and 1 are expected to be the same. [i](#)
- General: Further output of Intersection::heads will be suppressed during this calculation. [i](#)
- Complement: Nonatomic expression expected at position 2 in Complement[{1, 2, 3, 4, 5, 6, 7, 8}, 1]. [i](#)

Es bipartito: W1=1 y W2=Complement[{1, 2, 3, 4, 5, 6, 7, 8}, 1]

No es bipartito completo

In[216]:=

```
WEjer2GIIG5={1,2,3,4};
FEjer2GIIG5={{1,2},{2,3},{3,1},{1,4},{2,4},{3,4}};
AEjer2GII5a=MATRIZADYACENCIA[WEjer2GIIG5,FEjer2GIIG5];
MatrixForm[AEjer2GII5a]
REGULAR2[WEjer2GIIG5,FEjer2GIIG5]
COMPLETO[AEjer2GII5a]
BIPARTITO[WEjer2GIIG5,FEjer2GIIG5]
```

Out[219]//MatrixForm=

$$\begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

El grafo es 3-regular

Out[221]=

True

Intersection: Nonatomic expression expected at position 2 in $\{1, 2\} \cap 1$. [i](#)

Intersection: Nonatomic expression expected at position 1 in $1 \cap \{1, 2\}$. [i](#)

Intersection: Nonatomic expression expected at position 2 in $\{2, 3\} \cap 1$. [i](#)

General: Further output of Intersection::normal will be suppressed during this calculation. [i](#)

Complement: Nonatomic expression expected at position 2 in Complement[{1, 2, 3, 4}, 1]. [i](#)

Intersection: Heads Complement and List at positions 2 and 1 are expected to be the same. [i](#)

Intersection: Heads List and Complement at positions 2 and 1 are expected to be the same. [i](#)

Intersection: Heads Complement and List at positions 2 and 1 are expected to be the same. [i](#)

General: Further output of Intersection::heads will be suppressed during this calculation. [i](#)

Complement: Nonatomic expression expected at position 2 in Complement[{1, 2, 3, 4}, 1]. [i](#)

Es bipartito: W1=1 y W2=Complement[{1, 2, 3, 4}, 1]

No es bipartito completo

Ejercicio 2: Grafos II

Estudiar si el grafo del ejercicio 3 de la convocatoria ordinaria 2 del curso 2023/24, es regular, completo, bipartito y/o bipartito completo. Análogamente para un decágono.



In[223]:=

```
WEjer2GII={1,2,3,4,5,6};
FEjer2GII={1→2,2→3,3→4,4→5,5→6,6→1,1→4,2→5,3→6};
AEjer2GII=MATRIZADYACENCIA[WEjer2GII,FEjer2GII];
MatrixForm[AEjer2GII]
REGULAR[AEjer2GII]
COMPLETO[AEjer2GII]
BIPARTITO[WEjer2GII,FEjer2GII]
```

Out[226]//MatrixForm=

$$\begin{pmatrix} 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

Out[227]=

True

Out[228]=

False

- Intersection: Nonatomic expression expected at position 2 in {1, 2} ∩ 1. [i](#)
- Intersection: Nonatomic expression expected at position 1 in 1 ∩ {1, 2}. [i](#)
- Intersection: Nonatomic expression expected at position 2 in {2, 3} ∩ 1. [i](#)
- General: Further output of Intersection::normal will be suppressed during this calculation. [i](#)
- Complement: Nonatomic expression expected at position 2 in Complement[{1, 2, 3, 4, 5, 6}, 1]. [i](#)
- Intersection: Heads Complement and List at positions 2 and 1 are expected to be the same. [i](#)
- Intersection: Heads List and Complement at positions 2 and 1 are expected to be the same. [i](#)
- Intersection: Heads Complement and List at positions 2 and 1 are expected to be the same. [i](#)
- General: Further output of Intersection::heads will be suppressed during this calculation. [i](#)
- Complement: Nonatomic expression expected at position 2 in Complement[{1, 2, 3, 4, 5, 6}, 1]. [i](#)

Es bipartito: W1=1 y W2=Complement[{1, 2, 3, 4, 5, 6}, 1]

No es bipartito completo

Práctica 5

Códigos

In[230]:=

```
(*Conexo*)
CONEXO[A_]:=Module[{i,j,B},
  B=Sum[MatrixPower[A,j],{j,Dimensions[A][[1]]}];
  conexo=True;
  Do[
    Do[If[i≠j && B[[i,j]]==0,conexo=False];
      ,{i,Dimensions[B][[1]]}];
    ,{j,Dimensions[B][[1]]}];
```

```

conexo
]
(**)
MATRIZADYACENCIA[W_,F_]:=Module[{i,sparsematriz,CONTADORi,nuevoF},
nuevoF={};
Do[AppendTo[nuevoF,Position[W,F[[CONTADORi]][1]][1][1]→Position[W,F[[CONTADORi]][2]][1][1]];
,{CONTADORi,Length[F]}];
sparsematriz=SparseArray[Table[{nuevoF[[i]][1],nuevoF[[i]][2]}→1,{i,Length[F]}],{Length[W],
Length[W]}];
matrizadyacencia=sparsematriz+Transpose[sparsematriz];
Normal[matrizadyacencia]
];
(**)
COMPONENTESCONEXAS[matrizadyacencia_]:=Module[{j,B,conjunto,temp},
B=Sum[MatrixPower[matrizadyacencia,j],{j,Dimensions[matrizadyacencia][1]}];
componentesconexas={};
conjunto=Table[i,{i,Dimensions[B][1]}];
While[conjunto≠{},
temp={conjunto[[1]}];
Do[
If[B[[conjunto[[1]],conjunto[[i]]]≠0,AppendTo[temp,conjunto[[i]]];];
,{i,2,Length[conjunto]}];
AppendTo[componentesconexas,temp];
conjunto=Complement[conjunto,temp];
];
componentesconexas
]
(**)
EULER[matrizadyacencia_]:=Module[{A,i},
Euler=True;
If[CONEXO[matrizadyacencia],
A=matrizadyacencia.matrizadyacencia;
Do[
If[Mod[A[[i,i]],2]==1,Euler=False;Break[]];
,{i,Length[matrizadyacencia]}];
,
Euler=False;
];
Euler
]
(**)
HAMILTON[A_]:=Module[{ciclos,adyacentes,ultvertice,ciclostemp,k,t,j,i},
ciclos={1};
Do[
ciclostemp={};
Do[
adyacentes={};

ultvertice=ciclos[[k]][i];
Do[If[A[[ultvertice,j]]==1,AppendTo[adyacentes,j];,{j,Dimensions[A][1]}];

adyacentes=Complement[adyacentes,ciclos[[k]]];
If[adyacentes≠{},

Do[AppendTo[ciclostemp,Append[ciclos[[k]],adyacentes[[t]]];,{t,Length[adyacentes]}];

```

```

];
,{k,Length[ciclos]}}];
ciclos=ciclostemp;
,{i,Dimensions[A][[1]]-1}}];
ciclostemp={};
Do[If[A[[1,ciclos[[k]][Dimensions[A][[1]]]]==1,AppendTo[ciclostemp,Append[ciclos[[k]],1]]];,
{k,Length[ciclos]}}];
ciclostemp
];
(**)
distancia[u_,v_,matrizadyacencia_]:=Module[{B},
n=1;
B=matrizadyacencia;
While[B[[u,v]]==0 && n<Dimensions[B][[1]],n++;B=MatrixPower[matrizadyacencia,n]];
If[n==Dimensions[B][[1]],Print["No están conectados"]];
Print[B[[u,v]], " geodésica/s. Distancia: ",n];];
]
(**)
CAMINOS[vertice1_,vertice2_,longitud_,matrizadyacencia_]:=Module[{i,ultvertice,caminos,
adyacentes,caminostemp,k,j,t},
caminos={{vertice1}}];
Do[
caminostemp={};
Do[
adyacentes={};

ultvertice=caminos[[k]][[i]];
Do[If[matrizadyacencia[[ultvertice,j]]==1,AppendTo[adyacentes,j]]];,{j,Dimensions
[matrizadyacencia][[1]]}];

If[adyacentes!={},

Do[AppendTo[caminostemp,Append[caminos[[k]],adyacentes[[t]]]]];,{t,Length[adyacentes]}}];
];
,{k,Length[caminos]}}];
caminos=caminostemp;
,{i,longitud}}];
caminostemp={};
Do[If[caminos[[k]][[longitud+1]]==vertice2,AppendTo[caminostemp,caminos[[k]]]]];,{k,
Length[caminos]}}];
caminostemp
];
(**)
GEODESICAS[u_,v_,matrizadyacencia_]:=Module[{B},
n=1;
B=matrizadyacencia;
While[B[[u,v]]==0 && n<Dimensions[B][[1]],n++;B=MatrixPower[matrizadyacencia,n]];
If[n==Dimensions[B][[1]],Print["No están conectados"]];
Print[CAMINOS[u,v,n,matrizadyacencia]]];
]
(**)

```

Ejercicio 1

a) Cada uno de los grafos del ejercicio 5.29 del manual de practicas

b) Cada uno de los grafos del ejercicio 6.8 del manual de practicas

a) $G_1 = (W_1, F_1)$ con $W_1 = \{1, 2, 3, 4, 5, 6\}$ y $F_1 = \{1 \rightarrow 2, 3 \rightarrow 1, 4 \rightarrow 5, 6 \rightarrow 1\}$.

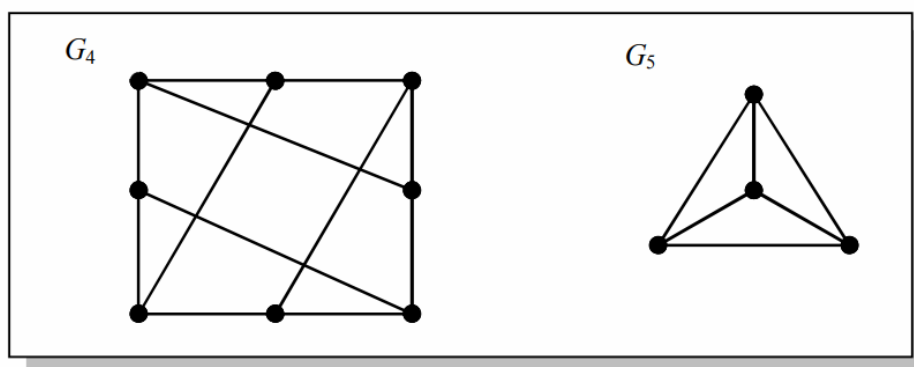
b) G_2 con matriz de adyacencia:

$$\begin{pmatrix} 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

c) G_3 con matriz de incidencia:

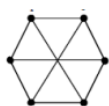
$$\begin{pmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \end{pmatrix}$$

d) Los grafos de la siguiente ilustración:



c) Un decágono

d) Grafo del ejercicio 3 de la convocatoria ordinaria 2 del curso 2023/24



In[238]:=

```

WG1={2,6,8,0,, "m", "a", "r", "t", "i", "n", "l", "u", "s"};
FG1={
  {0, "a"}, {0, "i"}, {0, "u"},
  {2, "a"}, {2, "i"}, {2, "u"},
  {6, "a"}, {6, "i"}, {6, "u"},
  {8, "a"}, {8, "i"}, {8, "u"},
  {2,6}, {2,8}, {2,6}
};
WG2={2,6,8,0,, "m", "a", "r", "t", "i", "n", "l", "u", "s"};
FG2={
  {"a", "m"}, {"a", "r"}, {"a", "t"}, {"a", "n"}, {"a", "l"}, {"a", "s"},
  {"i", "m"}, {"i", "r"}, {"i", "t"}, {"i", "n"}, {"i", "l"}, {"i", "s"},
  {"u", "m"}, {"u", "r"}, {"u", "t"}, {"u", "n"}, {"u", "l"}, {"u", "s"}
};
WG3={1,2,3,4,5,6};
FG3={1→2,3→1,4→5,6→1};
WG4={1,2,3,4};
FG4={2→1,4→1,1→2,3→2,2→3,4→3,1→4,3→4};
WG5={1,2,3,4,5};
FG5={1→4,1→5,2→4,2→5,3→4,3→5};
WG6={1,2,3,4,5,6,7,8};
FG6={1→2,2→3,3→4,4→5,5→6,6→7,7→8,8→1,1→4,2→7,3→6,5→8};
WG7={1,2,3,4};
FG7={1→2,2→3,3→1,1→4,2→4,3→4};
WG8={1,2,3,4,5,6,7,8,9,10};
FG8={{1,2},{2,3},{3,4},{4,5},{5,6},{6,7},{7,8},{8,9},{9,10},{10,1}};
WG9={1,2,3,4,5,6};
FG9={{1,2},{2,3},{3,4},{4,5},{5,6},{6,1},{1,4},{2,5},{3,6}};

```

Se pide:

i. Comprobar si es conexo

In[256]:=

```

"G1"
AG1=MATRIZADYACENCIA [WG1, FG1] ;
CONEXO [AG1]
"G2"
AG2=MATRIZADYACENCIA [WG2, FG2] ;
CONEXO [AG2]
"G3"
AG3=MATRIZADYACENCIA [WG3, FG3] ;
CONEXO [AG3]
"G4"
AG4=MATRIZADYACENCIA [WG4, FG4] ;
CONEXO [AG4]
"G5"
AG5=MATRIZADYACENCIA [WG5, FG5] ;
CONEXO [AG5]
"G6"
AG6=MATRIZADYACENCIA [WG6, FG6] ;
CONEXO [AG6]
"G7"
AG7=MATRIZADYACENCIA [WG7, FG7] ;
CONEXO [AG7]
"G8"
AG8=MATRIZADYACENCIA [WG8, FG8] ;
CONEXO [AG8]
"G9"
AG9=MATRIZADYACENCIA [WG9, FG9] ;
CONEXO [AG9]

```

Out[256]=

G1

Out[258]=

False

Out[259]=

G2

Out[261]=

False

Out[262]=

G3

Out[264]=

False

Out[265]=

G4

Out[267]=

True

Out[268]=

G5

Out[270]=

True

Out[271]=

G6

```

Out[273]=
  True

Out[274]=
  G7

Out[276]=
  True

Out[277]=
  G8

Out[279]=
  True

Out[280]=
  G9

Out[282]=
  True

```

ii. Calcular sus componentes conexas

```
In[283]:=
```

```

"G1"
COMPONENTESCONEXAS [AG1]
"G2"
COMPONENTESCONEXAS [AG2]
"G3"
COMPONENTESCONEXAS [AG3]
"G4"
COMPONENTESCONEXAS [AG4]
"G5"
COMPONENTESCONEXAS [AG5]
"G6"
COMPONENTESCONEXAS [AG6]
"G7"
COMPONENTESCONEXAS [AG7]
"G8"
COMPONENTESCONEXAS [AG8]
"G9"
COMPONENTESCONEXAS [AG9]

```

```

Out[283]=
  G1

Out[284]=
  {{1, 2, 3, 4, 7, 10, 13}, {5}, {6}, {8}, {9}, {11}, {12}, {14}}

Out[285]=
  G2

Out[286]=
  {{1}, {2}, {3}, {4}, {5}, {6, 7, 8, 9, 10, 11, 12, 13, 14}}

Out[287]=
  G3

Out[288]=
  {{1, 2, 3, 6}, {4, 5}}

Out[289]=
  G4

```

```

Out[290]=
  {{1, 2, 3, 4}}

Out[291]=
  G5

Out[292]=
  {{1, 2, 3, 4, 5}}

Out[293]=
  G6

Out[294]=
  {{1, 2, 3, 4, 5, 6, 7, 8}}

Out[295]=
  G7

Out[296]=
  {{1, 2, 3, 4}}

Out[297]=
  G8

Out[298]=
  {{1, 2, 3, 4, 5, 6, 7, 8, 9, 10}}

Out[299]=
  G9

Out[300]=
  {{1, 2, 3, 4, 5, 6}}

```

iii. Comprobar si es de Euler

```

In[301]:=
  "G1"
  EULER[AG1]
  "G2"
  EULER[AG2]
  "G3"
  EULER[AG3]
  "G4"
  EULER[AG4]
  "G5"
  EULER[AG5]
  "G6"
  EULER[AG6]
  "G7"
  EULER[AG7]
  "G8"
  EULER[AG8]
  "G9"
  EULER[AG9]

```

```

Out[301]=
  G1

Out[302]=
  False

Out[303]=
  G2

```

```
Out[304]=  
False
```

```
Out[305]=  
G3
```

```
Out[306]=  
False
```

```
Out[307]=  
G4
```

```
Out[308]=  
True
```

```
Out[309]=  
G5
```

```
Out[310]=  
False
```

```
Out[311]=  
G6
```

```
Out[312]=  
False
```

```
Out[313]=  
G7
```

```
Out[314]=  
False
```

```
Out[315]=  
G8
```

```
Out[316]=  
True
```

```
Out[317]=  
G9
```

```
Out[318]=  
False
```

iv. Comprobar si es de Hamilton

In[319]:=

```

"G1"
HAMILTON [AG1]
"G2"
HAMILTON [AG2]
"G3"
HAMILTON [AG3]
"G4"
HAMILTON [AG4]
"G5"
HAMILTON [AG5]
"G6"
HAMILTON [AG6]
"G7"
HAMILTON [AG7]
"G8"
HAMILTON [AG8]
"G9"
HAMILTON [AG9]

```

Out[319]=

G1

Out[320]=

{ }

Out[321]=

G2

Out[322]=

{ }

Out[323]=

G3

Out[324]=

{ }

Out[325]=

G4

Out[326]=

{ }

Out[327]=

G5

Out[328]=

{ }

Out[329]=

G6

Out[330]=

```

{ {1, 2, 3, 4, 5, 6, 7, 8, 1}, {1, 2, 3, 6, 7, 8, 5, 4, 1}, {1, 2, 7, 6, 3, 4, 5, 8, 1},
  {1, 2, 7, 8, 5, 6, 3, 4, 1}, {1, 4, 3, 2, 7, 6, 5, 8, 1}, {1, 4, 3, 6, 5, 8, 7, 2, 1},
  {1, 4, 5, 6, 3, 2, 7, 8, 1}, {1, 4, 5, 8, 7, 6, 3, 2, 1}, {1, 8, 5, 4, 3, 6, 7, 2, 1},
  {1, 8, 5, 6, 7, 2, 3, 4, 1}, {1, 8, 7, 2, 3, 6, 5, 4, 1}, {1, 8, 7, 6, 5, 4, 3, 2, 1} }

```

Out[331]=

G7

```

Out[332]=
{{ {1, 2, 3, 4, 1}, {1, 2, 4, 3, 1}, {1, 3, 2, 4, 1},
  {1, 3, 4, 2, 1}, {1, 4, 2, 3, 1}, {1, 4, 3, 2, 1} }}

Out[333]=
G8

Out[334]=
{{{ {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 1}, {1, 10, 9, 8, 7, 6, 5, 4, 3, 2, 1} }},
  {{ {G9, 2 G9, 3 G9, 4 G9, 5 G9, 6 G9, G9}, {G9, 2 G9, 3 G9, 6 G9, 5 G9, 4 G9, G9},
    {G9, 2 G9, 5 G9, 4 G9, 3 G9, 6 G9, G9}, {G9, 2 G9, 5 G9, 6 G9, 3 G9, 4 G9, G9},
    {G9, 4 G9, 3 G9, 2 G9, 5 G9, 6 G9, G9}, {G9, 4 G9, 3 G9, 6 G9, 5 G9, 2 G9, G9},
    {G9, 4 G9, 5 G9, 2 G9, 3 G9, 6 G9, G9}, {G9, 4 G9, 5 G9, 6 G9, 3 G9, 2 G9, G9},
    {G9, 6 G9, 3 G9, 2 G9, 5 G9, 4 G9, G9}, {G9, 6 G9, 3 G9, 4 G9, 5 G9, 2 G9, G9},
    {G9, 6 G9, 5 G9, 2 G9, 3 G9, 4 G9, G9}, {G9, 6 G9, 5 G9, 4 G9, 3 G9, 2 G9, G9} } } }

```

v. Utilizar el teorema del número de caminos para calcular la distancia entre el primer y último vértice de cada grafo

```

In[335]:=
"G1"
distancia[1,10,AG1]
"G2"
distancia[1,10,AG2]
"G3"
distancia[1,10,AG3]
"G4"
distancia[1,10,AG4]
"G5"
distancia[1,10,AG5]
"G6"
distancia[1,10,AG6]
"G7"
distancia[1,10,AG7]
"G8"
distancia[1,10,AG8]
"G9"
distancia[1,10,AG9]

```

```

Out[335]=
G1

1 geodésica/s. Distancia: 1

```

```

Out[337]=
G2

No están conectados

```

```

Out[339]=
G3

Part: Part 10 of {{0, 1, 1, 0, 0, 1}, {1, 0, 0, 0, 0, 0}, {1, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 1, 0}, {0, 0, 0, 1, 0, 0}, {1, 0, 0, 0, 0, 0}} does not exist. ⓘ

Part: Part 10 of {{0, 1, 1, 0, 0, 1}, {1, 0, 0, 0, 0, 0}, {1, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 1, 0}, {0, 0, 0, 1, 0, 0}, {1, 0, 0, 0, 0, 0}} does not exist. ⓘ

{{ {0, 1, 1, 0, 0, 1}, {1, 0, 0, 0, 0, 0}, {1, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 1, 0},
  {0, 0, 0, 1, 0, 0}, {1, 0, 0, 0, 0, 0} } [[1, 10]] geodésica/s. Distancia: 1

```

Out[341]=

G4

Part 10 of $\{\{0, 2, 0, 2\}, \{2, 0, 2, 0\}, \{0, 2, 0, 2\}, \{2, 0, 2, 0\}\}$ does not exist. [i](#)

Part 10 of $\{\{0, 2, 0, 2\}, \{2, 0, 2, 0\}, \{0, 2, 0, 2\}, \{2, 0, 2, 0\}\}$ does not exist. [i](#)

$\{\{0, 2, 0, 2\}, \{2, 0, 2, 0\}, \{0, 2, 0, 2\}, \{2, 0, 2, 0\}\} \llbracket 1, 10 \rrbracket$ geodésica/s. Distancia: 1

Out[343]=

G5

Part 10 of $\{\{0, 0, 0, 1, 1\}, \{0, 0, 0, 1, 1\}, \{0, 0, 0, 1, 1\}, \{1, 1, 1, 0, 0\}, \{1, 1, 1, 0, 0\}\}$ does not exist. [i](#)

Part 10 of $\{\{0, 0, 0, 1, 1\}, \{0, 0, 0, 1, 1\}, \{0, 0, 0, 1, 1\}, \{1, 1, 1, 0, 0\}, \{1, 1, 1, 0, 0\}\}$ does not exist. [i](#)

$\{\{0, 0, 0, 1, 1\}, \{0, 0, 0, 1, 1\}, \{0, 0, 0, 1, 1\}, \{1, 1, 1, 0, 0\}, \{1, 1, 1, 0, 0\}\} \llbracket 1, 10 \rrbracket$
geodésica/s. Distancia: 1

Out[345]=

G6

Part 10 of

$\{\{0, 1, 0, 1, 0, 0, 0, 1\}, \{1, 0, 1, 0, 0, 0, 1, 0\}, \{0, 1, 0, 1, 0, 1, 0, 0\}, \{1, 0, 1, 0, 1, 0, 0, 0\}, \{0, 0, 0, 1, 0, 1, 0, 1\}, \{0, 0, 1, 0, 1, 0, 1, 0\}, \{0, 1, 0, 0, 0, 1, 0, 1\}, \{1, 0, 0, 0, 1, 0, 1, 0\}\}$ does not exist. [i](#)

Part 10 of

$\{\{0, 1, 0, 1, 0, 0, 0, 1\}, \{1, 0, 1, 0, 0, 0, 1, 0\}, \{0, 1, 0, 1, 0, 1, 0, 0\}, \{1, 0, 1, 0, 1, 0, 0, 0\}, \{0, 0, 0, 1, 0, 1, 0, 1\}, \{0, 0, 1, 0, 1, 0, 1, 0\}, \{0, 1, 0, 0, 0, 1, 0, 1\}, \{1, 0, 0, 0, 1, 0, 1, 0\}\}$ does not exist. [i](#)

$\{\{0, 1, 0, 1, 0, 0, 0, 1\}, \{1, 0, 1, 0, 0, 0, 1, 0\}, \{0, 1, 0, 1, 0, 1, 0, 0\}, \{1, 0, 1, 0, 1, 0, 0, 0\}, \{0, 0, 0, 1, 0, 1, 0, 1\}, \{0, 0, 1, 0, 1, 0, 1, 0\}, \{0, 1, 0, 0, 0, 1, 0, 1\}, \{1, 0, 0, 0, 1, 0, 1, 0\}\} \llbracket 1, 10 \rrbracket$ geodésica/s. Distancia: 1

Out[347]=

G7

Part 10 of $\{\{0, 1, 1, 1\}, \{1, 0, 1, 1\}, \{1, 1, 0, 1\}, \{1, 1, 1, 0\}\}$ does not exist. [i](#)

Part 10 of $\{\{0, 1, 1, 1\}, \{1, 0, 1, 1\}, \{1, 1, 0, 1\}, \{1, 1, 1, 0\}\}$ does not exist. [i](#)

$\{\{0, 1, 1, 1\}, \{1, 0, 1, 1\}, \{1, 1, 0, 1\}, \{1, 1, 1, 0\}\} \llbracket 1, 10 \rrbracket$ geodésica/s. Distancia: 1

Out[349]=

G8

1 geodésica/s. Distancia: 1

Out[351]=

G9

Part 10 of $\{\{0, 1, 0, 1, 0, 1\}, \{1, 0, 1, 0, 1, 0\}, \{0, 1, 0, 1, 0, 1\}, \{1, 0, 1, 0, 1, 0\}, \{0, 1, 0, 1, 0, 1\}, \{1, 0, 1, 0, 1, 0\}\}$ does not exist. [i](#)

Part 10 of $\{\{0, 1, 0, 1, 0, 1\}, \{1, 0, 1, 0, 1, 0\}, \{0, 1, 0, 1, 0, 1\}, \{1, 0, 1, 0, 1, 0\}, \{0, 1, 0, 1, 0, 1\}, \{1, 0, 1, 0, 1, 0\}\}$ does not exist. [i](#)

$\{\{0, 1, 0, 1, 0, 1\}, \{1, 0, 1, 0, 1, 0\}, \{0, 1, 0, 1, 0, 1\}, \{1, 0, 1, 0, 1, 0\}, \{0, 1, 0, 1, 0, 1\}, \{1, 0, 1, 0, 1, 0\}\} \llbracket 1, 10 \rrbracket$ geodésica/s. Distancia: 1

$\{\{0, 1, 0, 1, 0, 1\}, \{1, 0, 1, 0, 1, 0\}, \{0, 1, 0, 1, 0, 1\}, \{1, 0, 1, 0, 1, 0\}, \{0, 1, 0, 1, 0, 1\}, \{1, 0, 1, 0, 1, 0\}\} \llbracket 1, 10 \rrbracket$ geodésica/s. Distancia: 1

vi. Determinar, si existen, un ciclo de Euler, un ciclo de Hamilton y una geodésica entre el primer y último vértice de cada grafo

In[353]:=

```

"G1"
GEODESICAS [1,13,AG1]
"G2"
GEODESICAS [1,13,AG2]
"G3"
GEODESICAS [1,6,AG3]
"G4"
GEODESICAS [1,2,AG4]
"G5"
GEODESICAS [1,5,AG5]
"G6"
GEODESICAS [1,8,AG6]
"G7"
GEODESICAS [1,4,AG7]
"G8"
GEODESICAS [1,10,AG8]
"G9"
GEODESICAS [1,6,AG9]

```

Out[353]=

G1

{ {1, 13} }

Out[355]=

G2

No están conectados

Out[357]=

G3

{ {1, 6} }

Out[359]=

G4

{ }

Out[361]=

G5

{ {1, 5} }

Out[363]=

G6

{ {1, 8} }

Out[365]=

G7

{ {1, 4} }

Out[367]=

G8

{ {1, 10} }

Out[369]=

G9

{ {1, 6} }

Práctica 6

Código

In[371]:=

```
MATRIZADYACENCIA[W_,F_]:=Module[{k,i,j,CONTADORi,nuevoF},
nuevoF={};
Do[AppendTo[nuevoF,Position[W,F[[CONTADORi]][1]][1]]->Position[W,F[[CONTADORi]][2]][1]][1]];
,{CONTADORi,Length[F]};
matrizadyacencia=Table[0,{i,Length[W]},{j,Length[W]};
Do[matrizadyacencia[[nuevoF[[k,1]],nuevoF[[k,2]]]=1;
matrizadyacencia[[nuevoF[[k,2]],nuevoF[[k,1]]]=1;
,{k,Length[F]};
matrizadyacencia];
```

Ejercicio 1.

Dados los siguientes grafos:

- Cada uno de los grafos del ejercicio 5.29 del manual de prácticas
- Cada uno de los grafos del ejercicio 6.8 del manual de prácticas
- Un decágono
- Grafo del ejercicio 3 de la convocatoria ordinaria 2 del curso 2023/24

In[372]:=

```

WG1={2,6,8,0,, "m", "a", "r", "t", "i", "n", "l", "u", "s"};
FG1={
  {0, "a"}, {0, "i"}, {0, "u"},
  {2, "a"}, {2, "i"}, {2, "u"},
  {6, "a"}, {6, "i"}, {6, "u"},
  {8, "a"}, {8, "i"}, {8, "u"},
  {2,6}, {2,8}, {2,6}
};
WG2={2,6,8,0,, "m", "a", "r", "t", "i", "n", "l", "u", "s"};
FG2={
  {"a", "m"}, {"a", "r"}, {"a", "t"}, {"a", "n"}, {"a", "l"}, {"a", "s"},
  {"i", "m"}, {"i", "r"}, {"i", "t"}, {"i", "n"}, {"i", "l"}, {"i", "s"},
  {"u", "m"}, {"u", "r"}, {"u", "t"}, {"u", "n"}, {"u", "l"}, {"u", "s"}
};
WG3={1,2,3,4,5,6};
FG3={1→2,3→1,4→5,6→1};
WG4={1,2,3,4};
FG4={2→1,4→1,1→2,3→2,2→3,4→3,1→4,3→4};
WG5={1,2,3,4,5};
FG5={1→4,1→5,2→4,2→5,3→4,3→5};
WG6={1,2,3,4,5,6,7,8};
FG6={1→2,2→3,3→4,4→5,5→6,6→7,7→8,8→1,1→4,2→7,3→6,5→8};
WG7={1,2,3,4};
FG7={1→2,2→3,3→1,1→4,2→4,3→4};
WG8={1,2,3,4,5,6,7,8,9,10};
FG8={{1,2},{2,3},{3,4},{4,5},{5,6},{6,7},{7,8},{8,9},{9,10},{10,1}};
WG9={1,2,3,4,5,6};
FG9={{1,2},{2,3},{3,4},{4,5},{5,6},{6,1},{1,4},{2,5},{3,6}};

```

Se pide:

- i. Calcular el número cromático y una coloración óptima

In[390]:=

```

NCOLORACIONES[matrizadyacencia_, lambda_] := Module[{coloracionestemp, t, h, i, j, colorady, k},
  coloraciones = Table[{k}, {k, lambda}];
  Do[
    coloracionestemp = {};
    Do[
      colorady = {};
      Do[
        If[matrizadyacencia[[t, i]] == 1
          , colorady = Union[colorady, {coloraciones[[h]] [[t]]}]];
        , {t, Length[coloraciones[[h]]}]];
      Do[
        If[Intersection[{j}, colorady] == {}, AppendTo[coloracionestemp, Append[coloraciones[[h]], j]]];
        , {j, lambda}];
      , {h, Length[coloraciones]};
      coloraciones = coloracionestemp;
      , {i, 2, Dimensions[matrizadyacencia] [[1]]};
      coloraciones
    ];

  NUMEROCROMATICO[matrizadyacencia_] := Module[{},
    t = 1;
    While[NCOLORACIONES[matrizadyacencia, t] == {}, t++];
    t];

  DIBUJAR[matrizadyacencia_] := Module[{colores, colorear, COLORES, i}, NUMEROCROMATICO
[matrizadyacencia]; colorear = coloraciones[[1]]; colores = {"Negro", "Rojo", "Verde",
"Azul", "Amarillo", "Cyan", "Marrón", "Naranja", "Púrpura", "Rosa", "Gris"}; COLORES =
{Black, Red, Green, Blue, Yellow, Cyan, Brown, Orange, Purple, Pink, Gray}; Print["Colores
asignados: "]; Do[Print["Vértice ", i, ", color ", colores[[colorear[[i]]]], {i,
Length[colorear]}]; GraphPlot[matrizadyacencia, VertexRenderingFunction -> ({
COLORES[[colorear[[#2]]], Disk[#1, 0.06`], Black, Text[#2, {#1[[1]] + .1, #1[[2]] + .1}]} &)]
]

```

In[393]:=

```

"G1"
AG1=MATRIZADYACENCIA [WG1, FG1] ;
NCOLORACIONES [AG1]
NUMEROCROMATICO [AG1]
DIBUJAR [AG1]
"G2"
AG2=MATRIZADYACENCIA [WG2, FG2] ;
NCOLORACIONES [AG2]
NUMEROCROMATICO [AG2]
DIBUJAR [AG2]
"G3"
AG3=MATRIZADYACENCIA [WG3, FG3] ;
NCOLORACIONES [AG3]
NUMEROCROMATICO [AG3]
DIBUJAR [AG3]
"G4"
AG4=MATRIZADYACENCIA [WG4, FG4] ;
NCOLORACIONES [AG4]
NUMEROCROMATICO [AG4]
DIBUJAR [AG4]
"G5"
AG5=MATRIZADYACENCIA [WG5, FG5] ;
NCOLORACIONES [AG5]
NUMEROCROMATICO [AG5]
DIBUJAR [AG5]
"G6"
AG6=MATRIZADYACENCIA [WG6, FG6] ;
NCOLORACIONES [AG6]
NUMEROCROMATICO [AG6]
DIBUJAR [AG6]
"G7"
AG7=MATRIZADYACENCIA [WG7, FG7] ;
NCOLORACIONES [AG7]
NUMEROCROMATICO [AG7]
DIBUJAR [AG7]
"G8"
AG8=MATRIZADYACENCIA [WG8, FG8] ;
NCOLORACIONES [AG8]
NUMEROCROMATICO [AG8]
DIBUJAR [AG8]
"G9"
AG9=MATRIZADYACENCIA [WG9, FG9] ;
NCOLORACIONES [AG9]
NUMEROCROMATICO [AG9]
DIBUJAR [AG9]

```

Out[393]=

G1

Out[395]=

```

NCOLORACIONES [
  {{0, 1, 1, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0}, {1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0},
  {1, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0}, {0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0},
  {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0},
  {1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0},
  {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0},
  {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0},
  {1, 1, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}}]

```

Out[396]=

3

Colores

asignados:

Vértice 1, color Negro

Vértice 2, color Rojo

Vértice 3, color Rojo

Vértice 4, color Negro

Vértice 5, color Negro

Vértice 6, color Negro

Vértice 7, color Verde

Vértice 8, color Negro

Vértice 9, color Negro

Vértice 10, color Verde

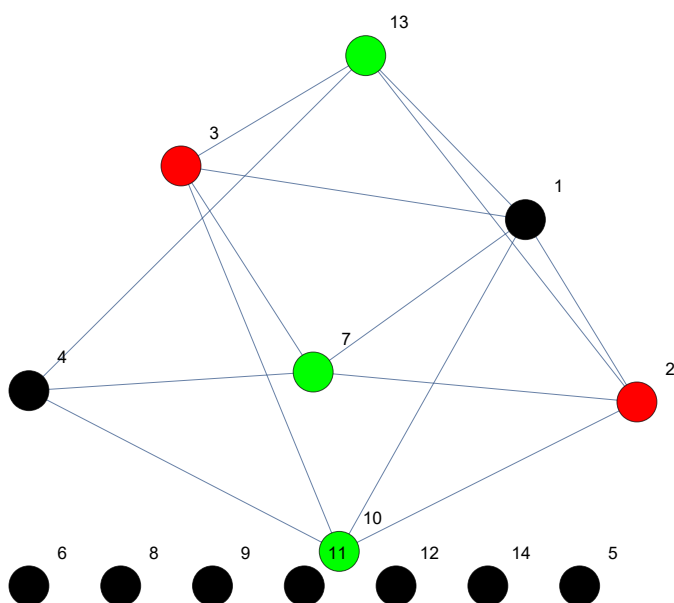
Vértice 11, color Negro

Vértice 12, color Negro

Vértice 13, color Verde

Vértice 14, color Negro

Out[397]=



Out[398]=

G2

Out[400]=

```
NCOLORACIONES [
  {{0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0},
  {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0},
  {0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0},
  {0, 0, 0, 0, 0, 1, 0, 1, 1, 0, 1, 1, 0, 1}, {0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0},
  {0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0}, {0, 0, 0, 0, 0, 1, 0, 1, 1, 0, 1, 1, 0, 1},
  {0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0}, {0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0},
  {0, 0, 0, 0, 0, 1, 0, 1, 1, 0, 1, 1, 0, 1}, {0, 0, 0, 0, 0, 0, 1, 0, 0, 1, 0, 0, 1, 0}}]
```

Out[401]=

2

Colores

asignados:

Vértice 1, color Negro

Vértice 2, color Negro

Vértice 3, color Negro

Vértice 4, color Negro

Vértice 5, color Negro

Vértice 6, color Negro

Vértice 7, color Rojo

Vértice 8, color Negro

Vértice 9, color Negro

Vértice 10, color Rojo

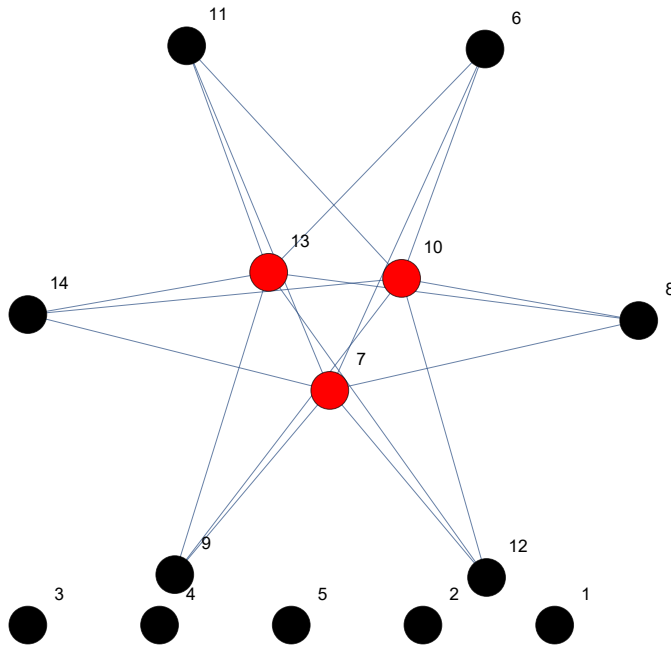
Vértice 11, color Negro

Vértice 12, color Negro

Vértice 13, color Rojo

Vértice 14, color Negro

Out[402]=



Out[403]=

G3

Out[405]=

```
NCOLORACIONES [ { {0, 1, 1, 0, 0, 1}, {1, 0, 0, 0, 0, 0},
  {1, 0, 0, 0, 0, 0}, {0, 0, 0, 0, 1, 0}, {0, 0, 0, 1, 0, 0}, {1, 0, 0, 0, 0, 0} } ]
```

Out[406]=

2

Colores

asignados:

Vértice 1, color Negro

Vértice 2, color Rojo

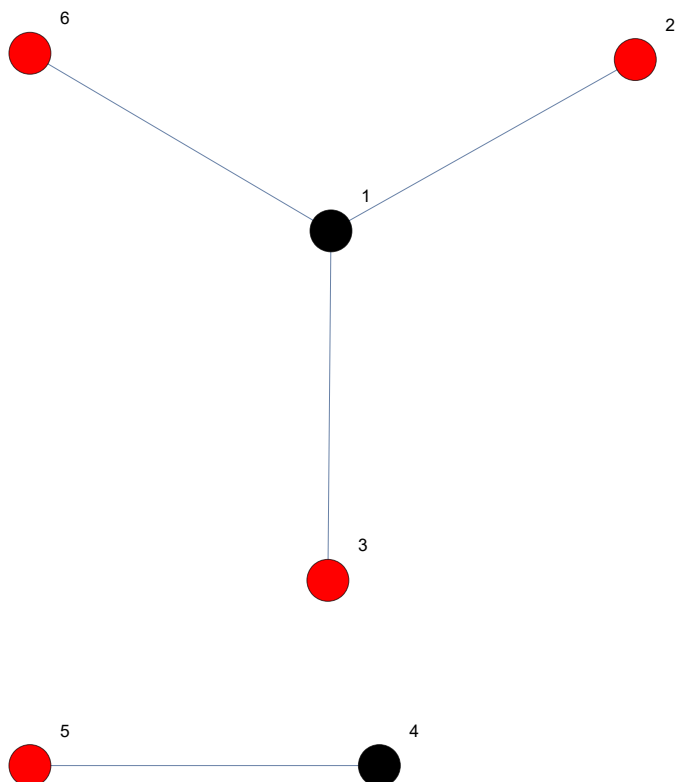
Vértice 3, color Rojo

Vértice 4, color Negro

Vértice 5, color Rojo

Vértice 6, color Rojo

Out[407]=



Out[408]=

G4

Out[410]=

NCOLORACIONES [{ { 0, 1, 0, 1 }, { 1, 0, 1, 0 }, { 0, 1, 0, 1 }, { 1, 0, 1, 0 } }]

Out[411]=

2

Colores

asignados:

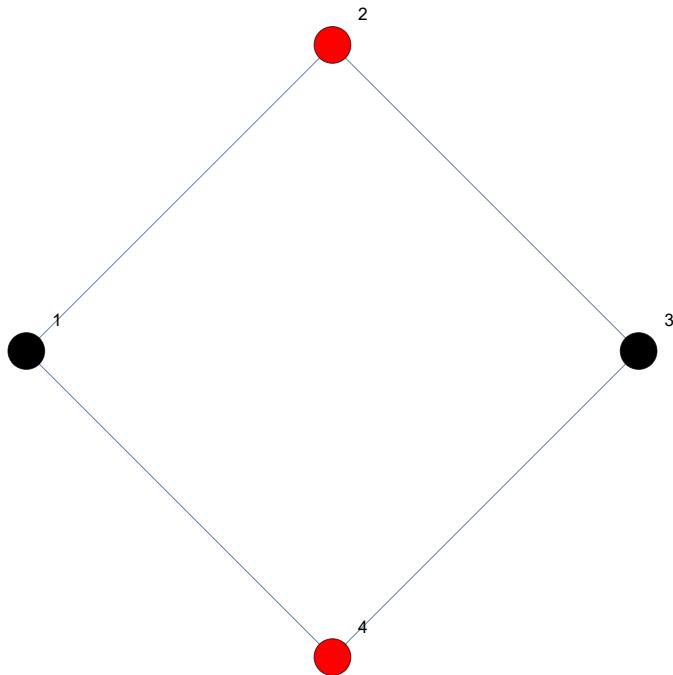
Vértice 1, color Negro

Vértice 2, color Rojo

Vértice 3, color Negro

Vértice 4, color Rojo

Out[412]=



Out[413]=

G5

Out[415]=

NCOLORACIONES [

{0, 0, 0, 1, 1}, {0, 0, 0, 1, 1}, {0, 0, 0, 1, 1}, {1, 1, 1, 0, 0}, {1, 1, 1, 0, 0}]

Out[416]=

2

Colores

asignados:

Vértice 1, color Negro

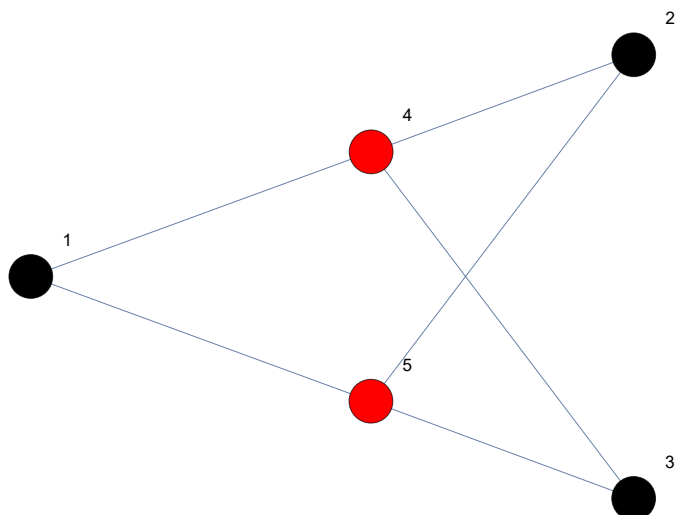
Vértice 2, color Negro

Vértice 3, color Negro

Vértice 4, color Rojo

Vértice 5, color Rojo

Out[417]=



Out[418]=

G6

Out[420]=

```

NCOLORACIONES [ { {0, 1, 0, 1, 0, 0, 0, 1}, {1, 0, 1, 0, 0, 0, 1, 0},
  {0, 1, 0, 1, 0, 1, 0, 0}, {1, 0, 1, 0, 1, 0, 0, 0}, {0, 0, 0, 1, 0, 1, 0, 1},
  {0, 0, 1, 0, 1, 0, 1, 0}, {0, 1, 0, 0, 0, 1, 0, 1}, {1, 0, 0, 0, 1, 0, 1, 0} } ]

```

Out[421]=

2

Colores

asignados:

Vértice 1, color Negro

Vértice 2, color Rojo

Vértice 3, color Negro

Vértice 4, color Rojo

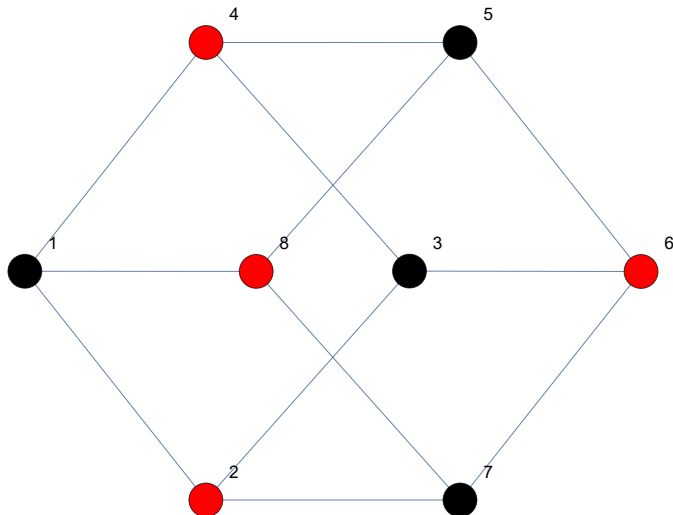
Vértice 5, color Negro

Vértice 6, color Rojo

Vértice 7, color Negro

Vértice 8, color Rojo

Out[422]=



Out[423]=

G7

Out[425]=

NCOLORACIONES [{ {0, 1, 1, 1}, {1, 0, 1, 1}, {1, 1, 0, 1}, {1, 1, 1, 0} }]

Out[426]=

4

Colores
asignados:

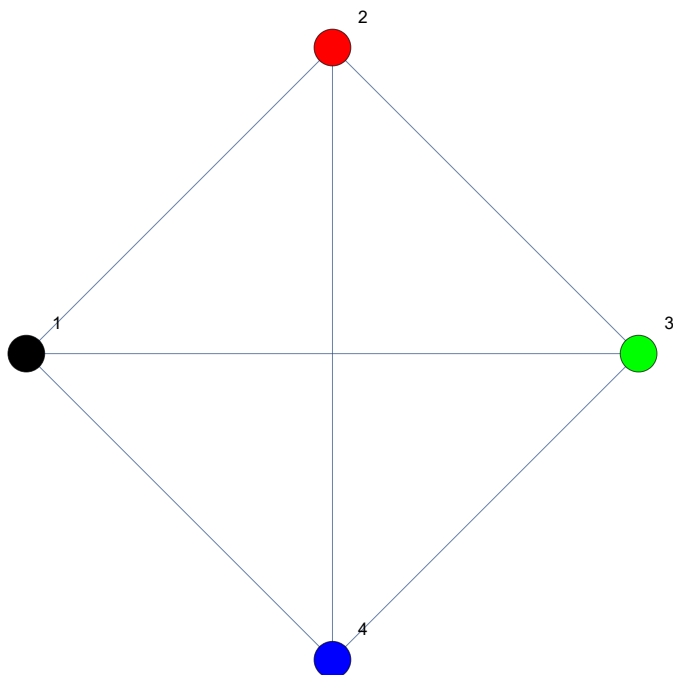
Vértice 1, color Negro

Vértice 2, color Rojo

Vértice 3, color Verde

Vértice 4, color Azul

Out[427]=



Out[428]=

G8

Out[430]=

```
NCOLORACIONES [ { {0, 1, 0, 0, 0, 0, 0, 0, 0, 1}, {1, 0, 1, 0, 0, 0, 0, 0, 0, 0},
  {0, 1, 0, 1, 0, 0, 0, 0, 0, 0}, {0, 0, 1, 0, 1, 0, 0, 0, 0, 0},
  {0, 0, 0, 1, 0, 1, 0, 0, 0, 0}, {0, 0, 0, 0, 1, 0, 1, 0, 0, 0},
  {0, 0, 0, 0, 0, 1, 0, 1, 0, 0}, {0, 0, 0, 0, 0, 0, 1, 0, 1, 0},
  {0, 0, 0, 0, 0, 0, 0, 1, 0, 1}, {1, 0, 0, 0, 0, 0, 0, 0, 1, 0} } ]
```

Out[431]=

2

Colores

asignados:

Vértice 1, color Negro

Vértice 2, color Rojo

Vértice 3, color Negro

Vértice 4, color Rojo

Vértice 5, color Negro

Vértice 6, color Rojo

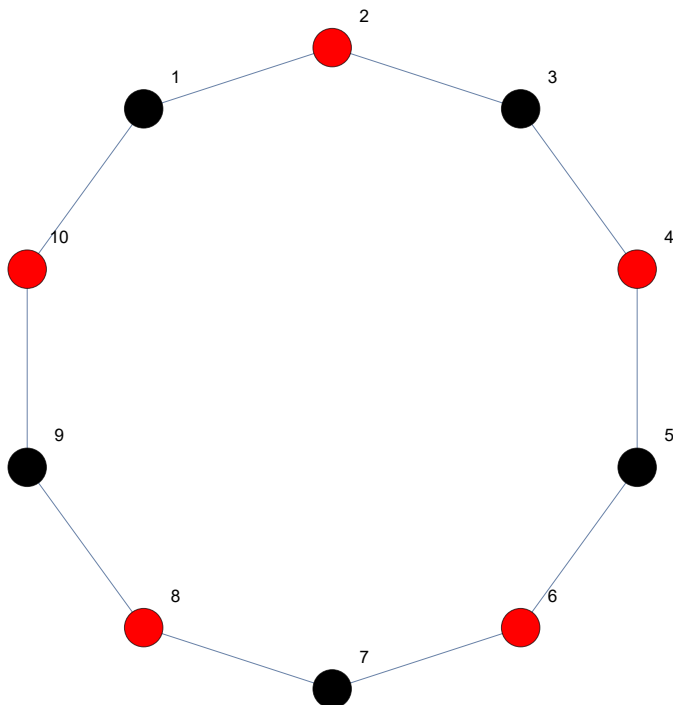
Vértice 7, color Negro

Vértice 8, color Rojo

Vértice 9, color Negro

Vértice 10, color Rojo

Out[432]=



Out[433]=

G9

```
Out[435]=
NCOLORACIONES[{{0, 1, 0, 1, 0, 1}, {1, 0, 1, 0, 1, 0},
{0, 1, 0, 1, 0, 1}, {1, 0, 1, 0, 1, 0}, {0, 1, 0, 1, 0, 1}, {1, 0, 1, 0, 1, 0}}]
```

```
Out[436]=
```

2

Colores

asignados:

Vértice 1, color Negro

Vértice 2, color Rojo

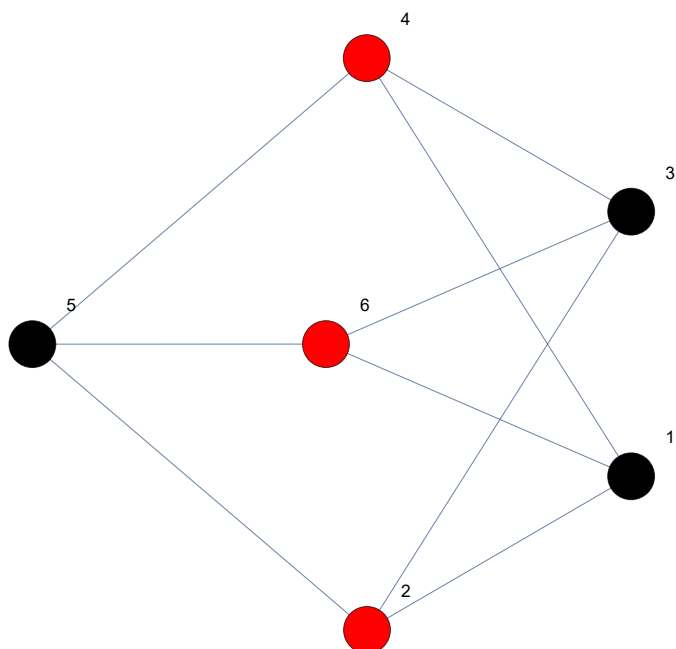
Vértice 3, color Negro

Vértice 4, color Rojo

Vértice 5, color Negro

Vértice 6, color Rojo

```
Out[437]=
```



ii. Determinar si es plano

```
In[444]:=
```

```
D3[matrizadyacencia_,i_,j_]:=Table[If[(k1==i && k2==j) || (k1==j && k2==i),1,matrizadyacencia
[[k1,k2]]],{k1,Dimensions[matrizadyacencia][[1]]},{k2,Dimensions[matrizadyacencia][[2]]}]
```

```
QUITARVERTICESGRADO210[matrizadyacencia_]:=Module[{grado2,t,w1,A,i,k,m,adya,k1,k2,k3,k4,
adyacentes,j},
```

```
Nueva=matrizadyacencia;
```

```
A=Nueva.Nueva;
```

```
grado2=True;t=1;
```

```
While[grado2,
```

```
w1={};
```

```
grado2=False;
```

```
If[Dimensions[A][[1]]>2,
```

```

Do[If[A[[i,i]]==2 || A[[i,i]]==1 || A[[i,i]]==0, AppendTo[W1,i];grado2=True;],{i,
Dimensions[A][[1]]}];
If[grado2,

Sort[W1];
Do[
m=W1[[Length[W1]-k+1]];

If[Dimensions[Nueva][[1]]>2,

If[A[[m,m]]==2,
adyacentes={};
Do[If[Nueva[[m,adya]]==1,AppendTo[adyacentes,adya]],{adya,Dimensions[Nueva][[1]]}];
Nueva=D3[Nueva,adyacentes[[1]],adyacentes[[2]]]
];
Nueva=Table[
If[k1>=m,k3=k1+1,k3=k1];
If[k2>=m,k4=k2+1,k4=k2];
Nueva[[k3,k4]]
,{k1,Dimensions[Nueva][[1]]-1},{k2,Dimensions[Nueva][[2]]-1}];

A=Nueva.Nueva;
];

,{k,Length[W1]}];
];

];t++;];
Nueva]

<<"Combinatorica`";
K33[W_,matrizadyacencia_]:=Module[{subconjuntos,nok33,i,k1,k2},
contiene=False;
subconjuntos=KSubsets[Complement[W,{W[[1]]}],2];
Do[
nok33=False;
A=Join[subconjuntos[[i]],{W[[1]]}];
B=Complement[W,A];
Do[
If[Sum[matrizadyacencia[[A[[k1]],B[[k2]]]],{k2,3}]<3,nok33=True;Break[]];
,{k1,3}];
If[!nok33,contiene=True;Break[]];
,{i,Length[subconjuntos]}];
contiene
]

HOMEOMORFOK5K33[matrizadyacencia_]:=Module[{Subconjuntos,i,j,cont1,matrizadyacenciaK5,
Nueva},
Nueva=QUITARVERTICESGRADO210[matrizadyacencia];
homeo=True;
If[Dimensions[Nueva][[1]]==6,
If[K33[{1,2,3,4,5,6},Nueva],homeo=False];];
matrizadyacenciaK5=Table[If[i==j,0,1],{i,5},{j,5}];
If[Dimensions[Nueva][[1]]==5,If[Nueva==matrizadyacenciaK5,homeo=False];];

```

```

homeo
]

PLANO[matrizadyacencia_]:=Module[{Nueva,k2,k3,subclados,lados},
Nueva=QUITARVERTICESGRADO210[matrizadyacencia];
tiempo=TimeUsed[];
plano=True;

vertices=Table[i,{i,Dimensions[Nueva][[1]]}];
lados={};

Do[
Do[
If[Nueva[[t1,t2]]==1,AppendTo[lados,t1->t2]];
,{t1,t2}];
,{t2,Dimensions[Nueva][[1]]}];

Do[
subclados=KSubsets[lados,k2];
Do[
If[HOMEOMORFOK5K33[MATRIZADYACENCIA[vertices,subclados[[k3]]]]==False,plano=False;subgrafo=
subclados[[k3]];Break[]];
If[TimeUsed[]-tiempo>5,Print[k2-9," ",Length[subclados]-k3];tiempo=TimeUsed[]];
,{k3,Length[subclados]}];
If[!plano,Break[]];
,{k2,Length[lados],9,-1}];

plano
]

```

 **General:** Combinatorica Graph and Permutations functionality has been superseded by preloaded functionality. The package now being loaded may conflict with this. Please see the Compatibility Guide for details.

In[450]:=

```

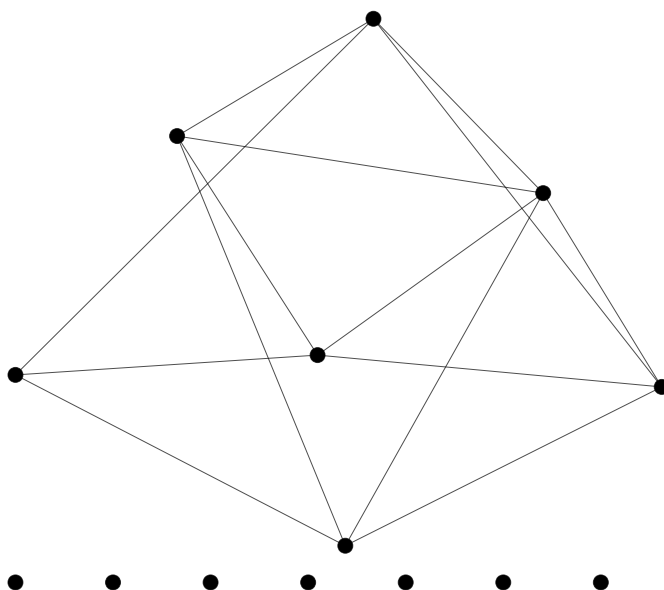
"G1"
GraphPlot [AG1,PlotStyle->{Black,PointSize[.05]}]
PLANO[AG1]
"G2"
GraphPlot [AG2,PlotStyle->{Black,PointSize[.05]}]
PLANO[AG2]
"G3"
GraphPlot [AG3,PlotStyle->{Black,PointSize[.05]}]
PLANO[AG3]
"G4"
GraphPlot [AG4,PlotStyle->{Black,PointSize[.05]}]
PLANO[AG4]
"G5"
GraphPlot [AG5,PlotStyle->{Black,PointSize[.05]}]
PLANO[AG5]
"G6"
GraphPlot [AG6,PlotStyle->{Black,PointSize[.05]}]
PLANO[AG6]
"G7"
GraphPlot [AG7,PlotStyle->{Black,PointSize[.05]}]
PLANO[AG7]
"G1"
GraphPlot [AG1,PlotStyle->{Black,PointSize[.05]}]
PLANO[AG1]
"G8"
GraphPlot [AG8,PlotStyle->{Black,PointSize[.05]}]
PLANO[AG8]
"G9"
GraphPlot [AG9,PlotStyle->{Black,PointSize[.05]}]
PLANO[AG9]

```

Out[450]=

G1

Out[451]=



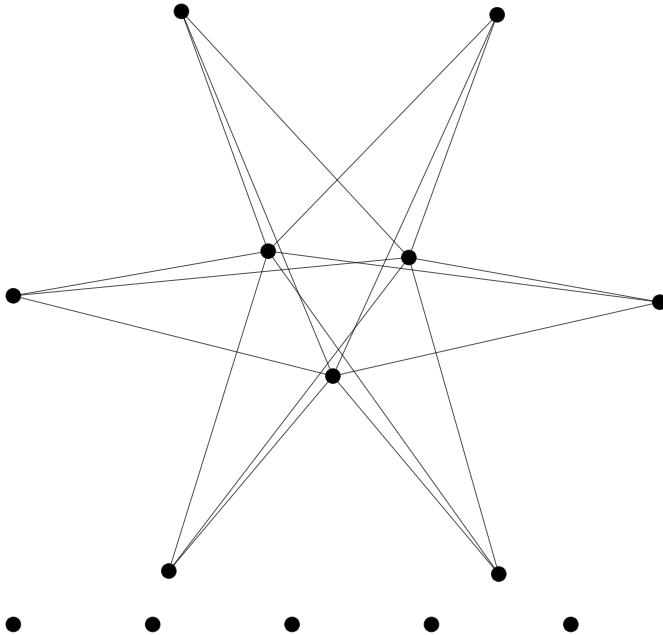
Out[452]=

False

Out[453]=

G2

Out[454]=



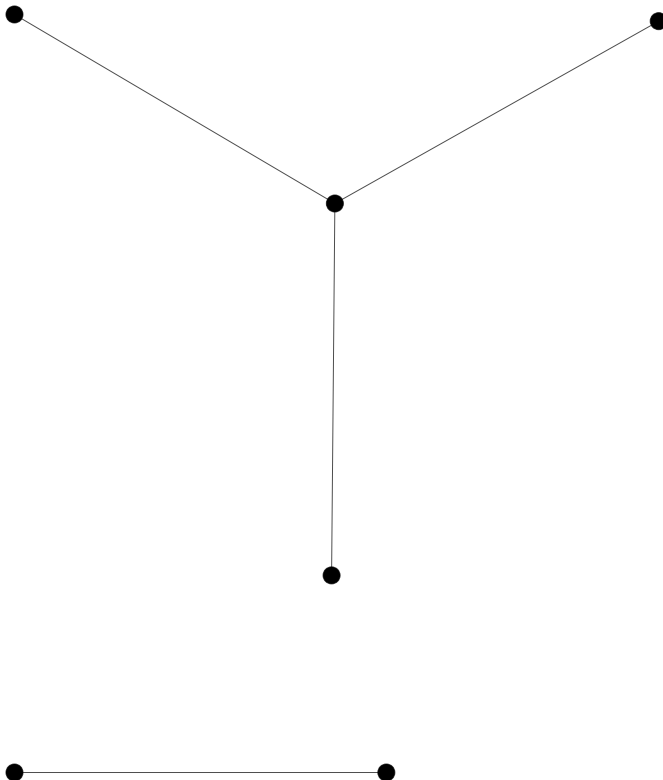
Out[455]=

False

Out[456]=

G3

Out[457]=



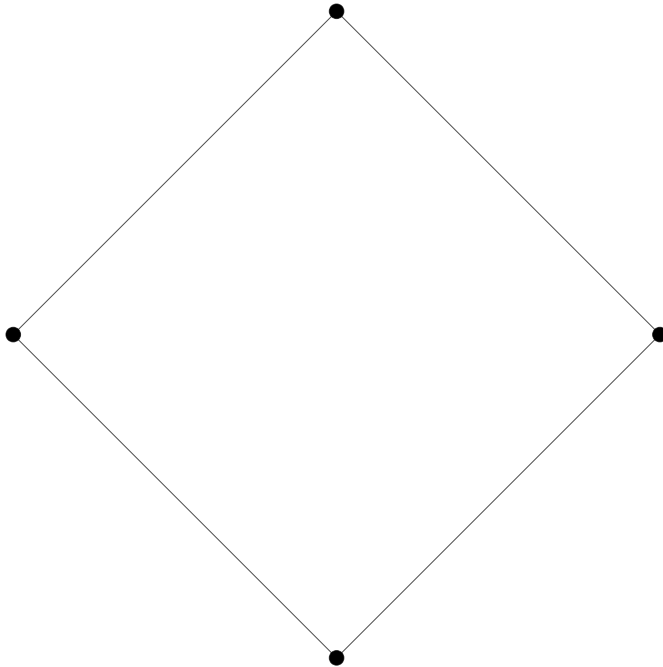
Out[458]=

True

Out[459]=

G4

Out[460]=



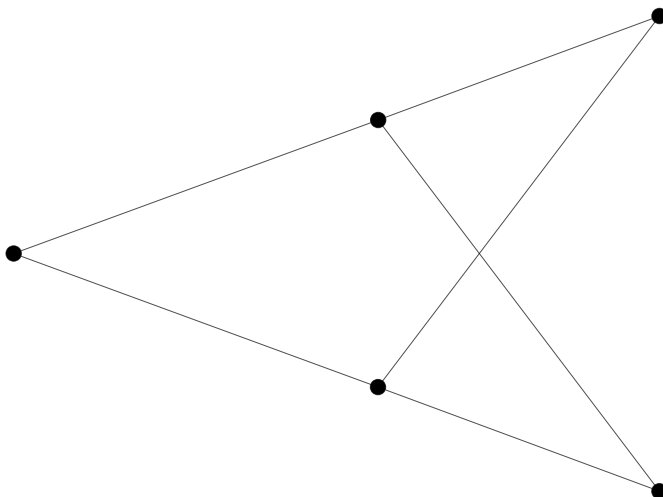
Out[461]=

True

Out[462]=

G5

Out[463]=



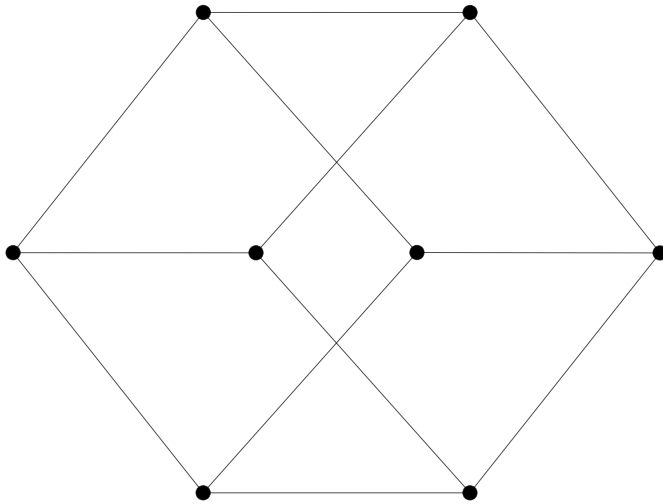
Out[464]=

True

Out[465]=

G6

Out[466]=



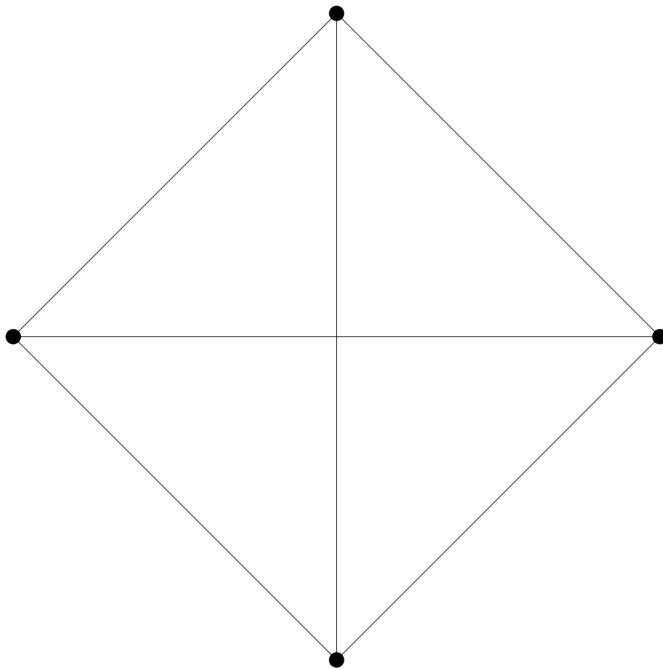
Out[467]=

True

Out[468]=

G7

Out[469]=



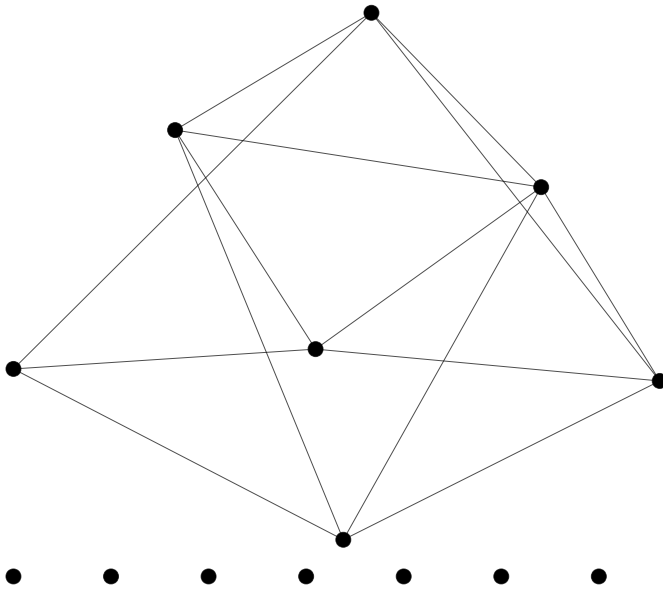
Out[470]=

True

Out[471]=

G1

Out[472]=



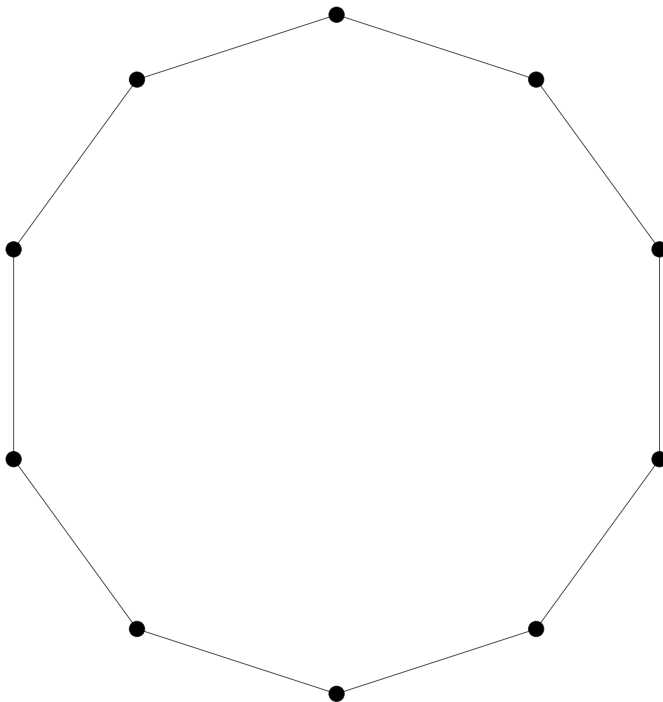
Out[473]=

False

Out[474]=

G8

Out[475]=



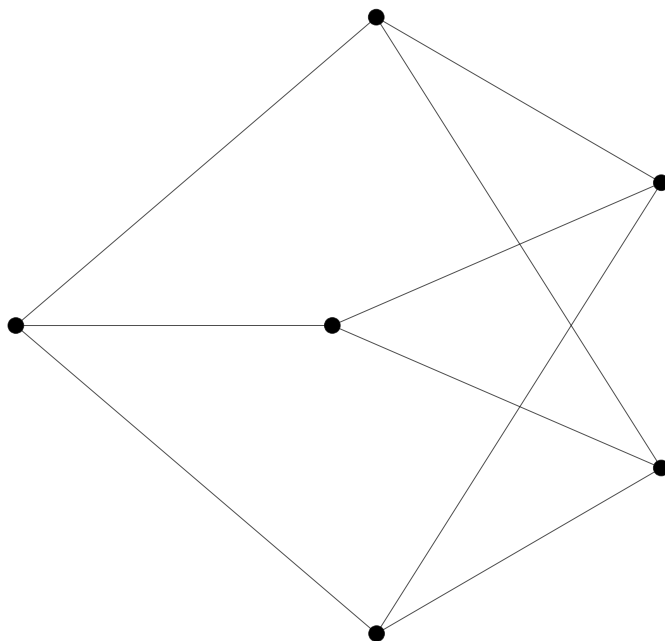
Out[476]=

True

Out[477]=

G9

Out[478]=



Out[479]=

False

iii. Comprobar si es un árbol

In[480]:=

```

CICLOSELEMENTALES[vertice1_, longitud_, A_] := Module[{i, ultvertice, caminos, adyacentes,
camino temp, k, j, t},
camino temp = {{vertice1}};
Do[
camino temp = {};
Do[
adyacentes = {};

ultvertice = camino temp[[k]][[i]];
Do[If[A[[ultvertice, j]] == 1, AppendTo[adyacentes, j]]], {j, Dimensions[A][[1]]};

adyacentes = Complement[adyacentes, camino temp[[k]]];
If[adyacentes == {},

Do[AppendTo[camino temp, Append[camino temp[[k]], adyacentes[[t]]]], {t, Length[adyacentes]}}];
, {k, Length[camino temp]};
camino temp = camino temp;
, {i, longitud - 1}];
camino temp = {};
Do[If[A[[camino temp[[k]][[longitud]], vertice1]] == 1, AppendTo[camino temp, Append[camino temp[[k],
vertice1]]]], {k, Length[camino temp]};
camino temp
];

BOSQUE[matrizadyacencia_] := Module[{n, longitud, i, },
bosque = True;
n = Dimensions[matrizadyacencia][[1]];

```

```

Do[
potencia=MatrixPower[matrizadyacencia,longitud];
Do[
If[potencia[[i,i]]>0,
If[CICLOSELEMENTALES[i,longitud,matrizadyacencia]≠{ },bosque=False;Break[]];
];
,{i,n}];
If[bosque=False,Break[]];
,{longitud,3,n}];
bosque
];

CONEXO[A_]:=Module[{i,j,B},
B=Sum[MatrixPower[A,j],{j,Dimensions[A][[1]]}];
conexo=True;
Do[
Do[If[i≠j && B[[i,j]]==0,conexo=False];
,{i,Dimensions[B][[1]]}];
,{j,Dimensions[B][[1]]}];
conexo
]

ARBOL[matrizadyacencia_]:=Module[{i,j,lados},
arbol=False;
If[CONEXO[matrizadyacencia],
arbol=(Dimensions[matrizadyacencia][[1]]==(Sum[matrizadyacencia[[i,j]],{i,Dimensions
[matrizadyacencia][[1]]},{j,Dimensions[matrizadyacencia][[2]]}]/2)+1);
];
arbol
]

COMPONENTESCONEXAS[matrizadyacencia_]:=Module[{j,B,conjunto,temp},
B=Sum[MatrixPower[matrizadyacencia,j],{j,Dimensions[matrizadyacencia][[1]]}];
componentesconexas={ };
conjunto=Table[i,{i,Dimensions[B][[1]]}];
While[conjunto≠{ },
temp={conjunto[[1]]};
Do[
If[B[conjunto[[1]],conjunto[[i]]]≠0,AppendTo[temp,conjunto[[i]]];
,{i,2,Length[conjunto]}];
AppendTo[componentesconexas,temp];
conjunto=Complement[conjunto,temp];
];
componentesconexas
]

IND2[W1_,matrizadyacencia_]:=Module[{n,i,j},
n=Length[W1];
matrizadyacencia2=Table[0,{i,n},{j,n}];
Do[Do[
matrizadyacencia2[[i,j]]=matrizadyacencia[[W1[[i]],W1[[j]]]];
,{i,n}];
,{j,n}];
matrizadyacencia2
];

```

```

BOSQUE2[matrizadyacencia_] := Module[{componentes, i},
bosque=True;
componentes=COMPONENTESCONEXAS[matrizadyacencia];
Do[
If[ARBOL[IND2[componentes[[i]],matrizadyacencia]]==False,bosque=False];
,{i,Length[componentes] }];
bosque]

```

In[487]:=

```
ARBOL[AG1]
```

Out[487]=

False

In[488]:=

```
ARBOL[AG2]
```

Out[488]=

False

In[489]:=

```
ARBOL [AG3]
"G4"
ARBOL [AG4]
"G5"
ARBOL [AG5]
"G6"
ARBOL [AG6]
"G7"
ARBOL [AG7]
"G8"
ARBOL [AG8]
"G9"
ARBOL [AG9]
```

Out[489]=

False

Out[490]=

G4

Out[491]=

False

Out[492]=

G5

Out[493]=

False

Out[494]=

G6

Out[495]=

False

Out[496]=

G7

Out[497]=

False

Out[498]=

G8

Out[499]=

False

Out[500]=

G9

Out[501]=

False